UNITED STATES OF AMERICA

BEFORE THE

FEDERAL ENERGY REGULATORY COMMISSION

Interconnection of Large Loads to the	)	Docket No. RM26-4-000
Interstate Transmission System	)	
	)	

SUPPLEMENTAL COMMENTS OF THE WORKING FOR ADVANCED TRANSMISSION TECHNOLOGIES COALITION

June 10, 2026

I. Introduction

The Working for Advanced Transmission Technologies (WATT) Coalition offers these supplemental comments in response to the Secretary of Energy's ("Secretary") proposed Advance Notice of Proposed Rulemaking (ANOPR) regarding the interconnection of large loads to the interstate transmission system in the above-docketed proceeding.

Our initial comments,[1] submitted with the Advancing Modern Powerlines (AMP) Coalition, defined and discussed the key value of Advanced Transmission Technologies ("ATTs") as the fastest way to increase transmission capacity to meet the needs of new

[1] Working for Advanced Transmission Technologies and Advancing Modern Powerlines Coalitions, Initial Comments, Docket No. RM 26-4-000 (filed Nov. 21, 2025).

load and cited many examples of the benefits of Dynamic Line Ratings ("DLR"), Advanced Power Flow Control ("APFC"), and Transmission Topology Optimization ("TTO") (the three canonical Grid Enhancing Technologies or "GETs") as well as High Performance Conductors ("HPCs"). Since our comment submission last fall, several deployments of ATTs have reinforced their value in load interconnection, prompting these supplemental comments. We discuss those case studies in Section II.

In initial comments, other commenters also noted the value of ATTs. Industrial Customers,[2] Southeast Public Interest Organizations,[3] ACORE,[4] and Corporate Energy Buyer's Association ("CEBA")[5] suggested that the Commission require consideration of ATTs in rules or actions related to large load interconnection. In Section III of these supplemental comments, we discuss why consideration in interconnection processes are important but likely not sufficient to unlock the 100GW+ of estimated available grid capacity with GETs, or the 100GW+ of capacity from other ATTs.[6]

To supplement that recommendation, the WATT Coalition has identified several options for action by the Commission that would:

1) Help regulators and stakeholders understand the potential benefits of ATTs (see Sections IV & VI);

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[2] Industrial Customer Organizations, Joint Reply Comments, Docket No. RM26-4-000 (filed Dec. 5, 2025).
[3] Southeast Public Interest Organizations, Initial Comments, Docket No. RM24-6-000 (filed Nov. 21, 2025).
[4] American Council on Renewable Energy, Initial Comments, Docket No. RM26-4-000 (filed Nov. 21, 2025).
[5] Corporate Energy Buyers Association, Initial Comments, Docket No. RM26-4-00 (filed Nov. 21, 2025).
[6] WATT Coalition and AMP Coalition, *Joint Response to DOE RFI on Accelerating Speed to Power/Winning the Artificial Intelligence Race,* at 2, (Nov. 21, 2025), https://watt-transmission.org/wp-content/uploads/2025/11/WATTAMP_DOE-Speed-to-power-RFI_responses_vFINAL.pdf.

2) Implement processes to increase transmission capacity in the near term to enable large load interconnection faster, while protecting ratepayers (see Section V).
3) Share learning to facilitate faster adoption of ATTs (see Section VII); and
4) Clarify incentive options for transmission owners deploying ATTs.

II. New examples of Grid Enhancing Technologies facilitating large load interconnection

Some leading American transmission owners are using GETs to manage new large load interconnections.

Georgia Power Company and Pacific Gas & Electric Corporation are demonstrating the value of APFC, with deployments that have unlocked more than 100 MW of capacity for new industrial and computing loads.[7] Georgia Power's first large-scale deployment uses 21 SmartValve modules installed across two 230kV circuits. Georgia Power also announced a further deployment expected to be live in 2028. PG&E's deployment is expected to be live by the end of 2026. The APFC devices in California will continue to be used after a transmission upgrade expected to come online in 2032, but with different

[7] Georgia Power, *Georgia Power Works with Smart Wires for First "SmartValve" Advanced Power Flow Control Technology* (Sept. 17, 2025), https://www.georgiapower.com/news-hub/press-releases/georgia-power-works-with-smart-wires-for-first-smartvalve.html; Smart Wires, *Smart Wires Partners with Georgia Power for First SmartValve Technology* (Sept. 16, 2025), https://www.smartwires.com/2025/09/16/smart-wires-partners-with-georgia-power-for-first-smartvalve-technology-to-increase-grid-efficiency-and-resiliency-as-energy-demand-grows/; PG&E and Smart Wires, *PG&E and Smart Wires Enhance Grid Reliability, Capacity for Data Centers in San Jose*, PR Newswire (May 28, 2025), https://www.smartwires.com/2025/05/28/pge-and-smart-wires-enhance-grid-reliability-capacity-for-data-centers-in-san-jose/ ("SmartValve devices deliver fast, high-impact upgrades that mitigate thermal overloads by up to 34%, enabling an additional 100 MW of firm power delivery over the existing lines.").

settings reflecting the changing grid topology, a benefit unique to APFC which will unlock even greater capacity.

Basin Electric Power Cooperative[8] has deployed DLR to help manage growth of oil and gas development in Northwest North Dakota, where load has more than doubled in the last 10 years and is forecasted to continue growing. Northwestern North Dakota had the most congested single transmission line in the Southwest Power Pool (SPP) footprint in 2023, with over $126 million of congestion costs between June and October alone, and earliest possible relief 18 months away. DLR was an immediate solution.

Basin Electric presented their initial DLR results at DTECH Midwest 2025, with findings between July and December 2024 of:

- Capacity above static rating 90% of the time.
- Avoided 500 hours of congestion (12% of period).
- Paid back full capital cost after just 38 days of operation.
- Over $20 million of realized savings from congestion in 2024 alone.

These examples show that the Commission action on GETs could unlock significant additional grid capacity for large loads. With Commission guidance and encouragement, many other transmission owners could unlock capacity comparable to these examples.

III. Requiring ATTs in load interconnection is valuable but not sufficient

[8] See slides on the "Ampacimon-Basin DLR Experience," presented by Jeremy Severson and Corey Hauser at DTECH Midwest 2025, July 14-16, 2025 in Minneapolis, MN, appended to these comments.

The record in this docket demonstrates the need to include ATTs in load interconnection processes, and that all decisions around planning and operating the grid have consequences for the speed of load interconnection and the cost of service.

The clearest application for GETs in this docket is in load interconnection studies themselves. We agree with the CEBA's recommendation "that the Commission require ATTs—including dynamic line ratings—to be evaluated alongside traditional upgrades for large load interconnections, consistent with Order No. 2023, given the significant benefits identified for consumers and interconnection customers."[9] ACORE similarly suggests that the Commission mirror Order No. 2023 and add DLR to the list of enumerated technologies.[10]

However, while requiring ATTs to be evaluated in interconnection processes is valuable, it is not sufficient to enable large load interconnection at the fastest reasonable pace and lowest possible cost. On how GETs could impact load interconnection in other processes, the Industrial Customer Organizations say:

> Paramount to these Reply Comments is our request that the Commission require evaluation and use of Grid Enhancing Technologies ("GETs") [...] in transmission studies where cost-effective. The time has come and gone for voluntary or incentivized consideration of GETs. It is well documented that use of GETs delivers cost reductions and cost avoidance to the benefit of consumers

[9] CEBA Initial Comments at 9.
[10] ACORE Initial Comments at 6.

> nationwide. Despite this, relatively few projects have been implemented. To strengthen the grid and protect consumers, the Commission must take a more aggressive posture, fully exercise its regulatory authority over the transmission of electricity in interstate commerce, and require GETs and advanced transmission technologies where cost-effective.[11]

Comments from the Data Center Coalition (DCC) also affirm that unlocking transmission capacity quickly will support load interconnection generally, and encourage the Commission to require GETs in "defined contexts" to accelerate adoption,[12] including moving ahead the rulemaking proceeding on DLR based on grid congestion.[13] In reply comments, DCC makes explicit that planning reform and GETs in generator interconnection processes would both support efficient load interconnection.[14] The WATT Coalition agrees with these large customer groups that utility adoption of GETs in many contexts will be useful for the overall goal of large load interconnection and achieving speed to power, in line with the ANOPR's stated aims.

There is some transmission owner support for GETs or ATTs requirements in the initial comments. National Grid states that, "By promoting efficient processes and supporting advanced technologies, the Commission can help modernize the grid while meeting urgent interconnection needs.".[15] American Transmission Company (ATC) "supports

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[11] Industrial Customer Organizations Joint Reply Comments at 2.
[12] Data Center Coalition, Initial Comments, Docket No. RM26-4-00, at 11 (filed Nov. 11, 2025).
[13] *Id.* at 10.
[14] Data Center Coalition, Reply Comments, Docket No. RM26-4-00, at 8 (filed Dec. 5, 2025).
[15] National Grid PLC, Initial Comments, Docket No. RM26-4-00, at 22 (filed Nov. 11, 2025).

leveraging these technologies as complementary measures within a comprehensive planning and operations framework—not as replacements for essential transmission infrastructure."[16] ATC argues that ATTs "can offer meaningful operational benefits, they are not substitutes for robust transmission expansion, particularly, given the size of the loads being interconnected, with some surpassing 1 GW."[17] The WATT Coalition agrees that large load interconnection will almost certainly require traditional transmission upgrades, but also that ATTs will likely be viable solutions for some elements of the necessary transmission capacity expansion. ATC says "These tools can optimize existing assets, improve flexibility, and support localized stability, but they cannot fully mitigate reliability risks posed by hybrid facilities and large, highly variable loads."[18] The WATT Coalition agrees that ATTs can provide these benefits and therefore should be considered in large load interconnection processes alongside traditional transmission system upgrades.

The PJM Transmission Owners are more strongly against GETs and ATTs, stating in their reply comments that "Transmission owners with relevant knowledge of their system conditions should be permitted to decide whether and where these technologies are employed. And their utility and purpose are remote from the primary goals of the ANOPR."[19] The WATT Coalition believes that the case studies submitted in our initial comments, along with comments submitted by ATC, National Grid, and others in this

[16] American Transmission Company, Reply Comments, Docket No. RM26-4-00, at 8 (filed Dec. 5, 2025).
[17] *Id.*
[18] *Id.*
[19] PJM Transmission Owners, Reply Comments, Docket No. RM26-4-00, at 8-9 (filed Dec. 5, 2025).

proceeding show that ATTs have significant value in the context of load interconnection. However, Dominion Energy is currently the only PJM transmission owner with significant DLR adoption planned in the next six months.[20] This suggests Commission action may be needed to take full advantage of the role of ATTs in facilitating large load interconnection.

Another relevant example is FirstEnergy Service Company (FirstEnergy), a transmission provider in PJM, among other regions, that opposes requirements for ATTs in this proceeding. FirstEnergy contends:

> Although transmission providers should consider GETs and ATTs in connection with large load interconnections, the Commission should not mandate the evaluation and use of GETs and ATTs. Layering such specific requirements on transmission providers will increase the burdens and complexity of compliance with any final rule and thus could frustrate the Department's "speed to power" goal.[21]

FirstEnergy further argues that, "As with automation and AI tools, transmission providers already look for opportunities to use GETs and ATTs."[22]

While the WATT Coalition is sensitive to the administrative burden of new requirements as well as the speed to power goals that underpin the ANOPR, we believe that the

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[20] PJM, *PJM Projected TO DLR Facilities List as of 5/1/2026*, https://www.pjm.com/-/media/DotCom/committees-groups/committees/oc/2026/20260507/20260507-item-10---dlr-update.pdf, accessed June 8, 2026.
[21] FirstEnergy Service Company, Reply Comments, Docket No. RM26-4-00, at 5 (filed Dec. 5, 2025).
[22] *Id.*

Commission can advance productive ATT policy, including requirements, without slowing progress on large load interconnection. Indeed, accelerated ATT adoption is more likely to speed the interconnection of new large loads to constrained transmission systems rather than frustrate speed to power goals. If transmission owners are already looking for opportunities to deploy ATTs, any requirements should be reasonably easy to implement. Ensuring that ATTs, as FirstEnergy suggests, are evaluated in interconnection efforts, as suggested by CEBA and others, and in other planning and operational processes, as raised by the Industrial Customer Organizations and DCC, is supported by record evidence and has a strong showing that the benefits far outweigh the burdens.

In addition to including ATTs in interconnection studies, the WATT Coalition has additional suggestions for FERC-jurisdictional policy that could accelerate the discovery of useable transmission capacity with GETs.

IV. Policy lever: require reporting on potential of GETs to address grid congestion and improve investment efficiency

The following suggestions are focused on reporting requirements, including: (1) requiring regional transmission organizations (RTOs) to study the potential for DLR to address congestion; (2) requiring RTOs to study the potential for other GETs to address congestion; and (3) requiring transmission owners to report on asset replacement and maintenance investments. The Commission has broad authority to direct reporting; reporting is generally a low-burden requirement; and the increased transparency from reporting on congestion and the potential for GETs to mitigate such congestion and

improve investment efficiency can inform state and federal policy around accelerated and targeted deployment. For example, the system-wide studies of DLR potential discussed under section IV.a. below should help the Commission set a threshold for requiring DLR systems on highly constrained lines, and thus finalizing the rule proposed in Docket No. RM24-6-000 on the Implementation of Dynamic Line Ratings.[23] The transparency envisioned by these reporting requirements is also critical to ensuring just and reasonable rates by identifying opportunities to maximize the use of our existing grid, to the benefit of ratepayers across the country, and to ensure our investments in new transmission are efficient and cost effective—key components of just and reasonable rates.

a. Require RTOs to study the potential for DLRs to address congestion

The fastest action that can be taken to identify potential untapped grid capacity is large-scale studies by RTOs. RTOs have sufficient data on grid constraints and electricity prices to use publicly available DLR datasets and/or software from the National Laboratory of the Rockies (NLR) to quickly determine the likely magnitude of value from DLR. This reporting would help transmission owners and stakeholders understand the potential value of DLR, and help the Commission determine if and how DLR should be required.[24] The RTOs could seek technical assistance from the NLR to undertake a

[23] Implementation of Dynamic Line Ratings, Advance Notice of Proposed Rulemaking, 187 FERC ¶ 61,024 (2024).

[24] The NLR has an open-source python package, which anyone can use to generate an approximate seven-year historical timeseries of hourly ratings for any lines in the contiguous United States. It uses the HIFLD dataset, combining 2 km wind speed/direction and air temperature/pressure from the WIND Toolkit with 4 km solar irradiance from the NSRDB. Their research is presented in "Hourly dynamic line ratings for existing transmission across the contiguous United States (preliminary results)," Patrick R. Brown, et al., October 2024, https://docs.nlr.gov/docs/fy25osti/91599.pdf.

study. RTOs should be able to complete these studies in a matter of months. The Commission may consider making this study a part of standard reporting. The Independent Market Monitors (IMMs) report on grid congestion in State of the Market Reports, and have recommended line rating methodology improvements to reduce congestion.[25] The MISO IMM found that for Winter 2023, implementing Ambient Adjust Ratings and Emergency Ratings could have saved $168 million in real-time congestion value.[26] Requiring RTOs to report on mitigation strategies is a useful next step.

As a starting point, the Commission could require all RTOs to assess the potential value of wind-based DLR to

1) Reduce energy prices through more efficient economic dispatch in the day-ahead and real-time markets; and

2) improve dispatch during extreme weather.

This study would support the Commission's next steps the open docket on the Implementation of Dynamic Line Ratings. To do this, the RTOs should first quantify expected savings based on implementing wind-based DLR across the whole system, or on only highly-constrained lines. The studies should include scenarios studying the last three events when the system was in Energy Emergency Alert Stage 1 (EEA1)[27], as well

[25] Grid Strategies LLC, *Transmission Congestion for 2024*, at 12 (Nov. 2025), https://gridstrategiesllc.com/wp-content/uploads/GS_Transmission-Congestion-for-2024.pdf.
[26] Potomac Economics, *IMM Quarterly Report: Winter 2023*, at 9 (Mar. 2023), https://cdn.misoenergy.org/20230321%20Markets%20Committee%20of%20the%20BOD%20Item%2006%20IMM%20Quarterly%20Report628268.pdf.
[27] The NERC-mandated alert when all available generation resources are in use.

as other representative cases from the past three years of grid operations. The Brattle Group's GETs modeling work reported in the *Unlocking the Queue with Grid Enhancing Technologies* Report provides an example of how to study congestion with select planning cases.[28] The studies should evaluate how DLR would have improved economic dispatch if deployed on:

1) All lines – which would identify whether the congestion cost savings from deploying DLR on constrained lines would offset the cost of deploying them on unconstrained lines as well. Those deployments on unconstrained lines would alert transmission owners to asset condition issues, provide useful data when making planning decisions, and potentially reduce future outage-driven congestion.

2) Lines with >$500,000 average congestion over the last three years. This reflects a threshold proposed by the WATT Coalition two years ago, based on the expectation that these deployments would easily pay for themselves.[29]

3) Lines with >$2,00,000 average congestion over the last three years. This reflects a threshold proposed by PJM 2 years ago.[30]

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[28] The Brattle Group, *Unlocking the Queue with Grid Enhancing Technologies* (Feb. 2021), https://watt-transmission.org/wp-content/uploads/2021/02/Brattle__Unlocking-the-Queue-with-Grid-Enhancing-Technologies__Final-Report_Public-Version.pdf90.pdf.

[29] Technology and Clean Energy Trade Associations, Initial Comments, Docket No. RM24-6-00, at 11 (filed Oct. 15, 2024), https://watt-transmission.org/wp-content/uploads/2024/10/INITIAL-COMMENTS-OF-THE-TECHNOLOGY-AND-CLEAN-ENERGY-TRADE-ASSOCIATIONS.pdf.

[30] PJM, Initial Comments, Docket No. AD22-5-000, at 9 (filed May 9, 2022), https://www.pjm.com/-/media/DotCom/documents/ferc/filings/2022/20220509-ad22-5-000.ashx.

4) And all lines with projected/modeled future congestion. While production cost modeling grossly under-forecasts congestion,[31] it does still identify some economic constraints. For those constraints that do show up in models, DLR should be deployed if it is anticipated to be cost effective.

These savings across the last three years, and in EEA1 scenarios, should be compared to expected costs for DLR implementation (which Midcontinent Independent System Operator (MISO) Transmission Owners estimated in 2021 as $100,000-$200,000 per line.[32]

These RTO studies would help identify tradeoffs between different potential thresholds for DLR deployment requirements, and how much congestion is captured in grid planning modeling when choosing to deploy these technologies versus how congestion emerges in operations.

b. Require RTOs to study other GETs for congestion management

Similar to the study of DLR described above, the Commission could require RTOs to study transmission topology optimization to reduce or resolve constraints where DLR is

[31] E.g. MISO's production cost modeling approach only identified any congestion expectation on one of the top ten most congested elements on the system. See *Update on MTEP23 Near-Term Congestion Study,* presented to the MISO Planning Subcommittee September 27, 2023. Accessed June 9 2026 at https://cdn.misoenergy.org/20230927%20PSC%20Item%2006%20Near%20Term%20Congestion%20Study%20(PAC-2021-1)630307.pdf.

[32] MISO Transmission Owners, Initial Comments, Docket No. RM20-16-000 (filed Mar. 22, 2021), at 47 (deriving $1.5 billion by estimating $150,000 per line multiplied by 10,000 lines on the MISO system).

not sufficient. It would be efficient for RTOs to use the same study cases for this second technology, which addresses congestion at the regional level by optimizing grid topology.

Evidence of the value of TTO has been presented to the Commission since 2013,[33] with regular updates at yearly FERC Software Conferences, and an accumulation of case studies of cost savings and reliability benefits in the interim.[34] SPP reports on many compelling case studies in the *Tariff Revisions to Implement Economic Topology*

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[33] *See* Pablo A. Ruiz et al., Transmission Topology Optimization, Presentation at Technical Conference on Increasing Market and Planning Efficiency through Improved Software, Docket No. AD10-12-003 (June 2013); Pablo A. Ruiz et al., Topology Control Algorithms Simulations in PJM with AC Modeling, Presentation at Technical Conference on Increasing Market and Planning Efficiency through Improved Software, Docket No. AD10-12-005 (June 24, 2014); Pablo A. Ruiz et al., Preserving Revenue Adequacy in FTR Markets with Changing Topology, Presentation at Technical Conference on Increasing Market and Planning Efficiency through Improved Software, Docket No. AD10-12-006 (June 24, 2015); Pablo A. Ruiz et al., Transmission Topology Optimization Impacts in Day-Ahead Markets — Simulations in PJM, Presentation at Technical Conference on Increasing Market and Planning Efficiency through Improved Software, Docket No. AD10-12-007 (June 28, 2016); Pablo A. Ruiz et al., Transmission Topology Optimization Applications to Enhance System Reliability and Resilience and to Increase Market and Planning Efficiency in Southwest Power Pool, Presentation at Technical Conference on Increasing Market and Planning Efficiency through Improved Software, Docket No. AD10-12-009 (June 26, 2018); Pablo A. Ruiz et al., Transmission Topology Optimization to Efficiently Mitigate Congestion and Overloads, Presentation at Technical Conference on Increasing Market and Planning Efficiency through Improved Software, Docket No. AD10-12-010 (June 2019); Pablo A. Ruiz, Transmission Topology Optimization Case Studies on Market Efficiency Improvements and Reliability and Resilience Enhancements, Presentation at Technical Conference on Increasing Market and Planning Efficiency through Improved Software, Docket No. AD10-12-011 (June 24, 2020); Pablo A. Ruiz and Xiao Li, Transmission Topology Optimization to Efficiently Mitigate Congestion & Overloads, Presentation at Technical Conference on Increasing Market and Planning Efficiency through Improved Software, Docket No. AD10-12-012 (June 23, 2021); Pablo A. Ruiz et al., Congestion and Overload Mitigation Using Optimal Transmission Reconfigurations — Experience in MISO and SPP, Presentation at Technical Conference on Increasing Market and Planning Efficiency through Improved Software, Docket No. AD10-12-013 (June 23, 2022); Pablo A. Ruiz, Presentation at Technical Conference on Increasing Market and Planning Efficiency through Improved Software, Docket No. AD10-12-014 (June 2023); Pablo A. Ruiz et al., Presentation at Technical Conference on Increasing Market and Planning Efficiency through Improved Software, Docket No. AD10-12-015 (July 2024); Pablo A. Ruiz et al., Presentation at Technical Conference on Increasing Market and Planning Efficiency through Improved Software, Docket No. AD10-12-016 (July 2025).

[34] See *Select Topology Optimization Case Studies*, NewGrid Inc., (January 11, 2026) https://newgridinc.com/wp-content/uploads/2026/01/topology-optimization-case-studies.pdf

Optimization, submitted May 21, 2026, which are appended to these comments. Examples include:

1. A TTO study in their footprint which found "historical system operating limit violations could be eliminated for 75% of the constraints analyzed. The resulting congestion cost savings in the real-time markets were estimated to be in the range of $18-44 million annually."[35]
2. A study of Evergy's footprint using four months of spring and summer 2022 operations data which found "reconfigurations that, had they been implemented, could have eliminated 98% of overloads and reduce congestion costs by 85% for congestion patterns associated with those constraints, including the impacts of additional constraints resulting from the change in flow patterns."[36]
3. Two transmission reconfigurations implemented during Winter Storm Elliot in 2022 which "released up to 845 MW from otherwise stranded generation under Energy Emergency Alert conditions, which supplied 14 GWh of additional energy during the event."[37]

SPP's filing discusses the extensive stakeholder discussion of the benefits of topology optimization. They note that the coincidence of technological advancements and increasing congestion costs as the reason for this change:

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[35] Southwest Power Pool, Submission of Tariff Revisions to Implement Economic Topology Optimization, Transmittal Letter, Docket No. ER26-2592-000 (filed May 21, 2026).
[36] *Id.* at 11
[37] *Id.* at 11

> The industry has found itself in a new age where technological advances, particularly grid-enhancing technologies ("GETs"), allow grid operators and experts to explore and embrace the "computational complexity" of topology optimization. By "topology optimization," SPP refers to alteration of a transmission network's configuration in a way that economically and reliably manages congestion. Such reconfiguration has the potential to reduce production and congestions costs. By way of example, from 2020 to 2021, SPP congestion costs increased from $450 million to $1.2 billion. GETs which allow for, say, topology reconfiguration have the potential to decrease costs such as these.[38]

They cite processes in place at MISO[39] and in development at ERCOT[40] to allow customers to request grid reconfigurations for economic efficiency. Study of the potential for TTO by all RTOs on a system-wide basis would accelerate progress towards implementation of TTO tools in all relevant system planning and operations processes.

c. Require transmission owners to report on asset replacement and maintenance investments

[38] *Id.* at 4

[39] MISO RSC Meeting, MISO Economic Reconfiguration Process Value and Suggestions for Further Improvements (Oct. 9, 2025), available at: https://cdn.misoenergy.org/20251009%20RSC%20Item%2011%20Reconfiguration%20for%20Cost%20Congestion%20Process%20Enhancements721553.pdf

[40] ERCOT, Grid Transformation – Technology Initiatives - Roadmaps, available at: https://www.ercot.com/files/docs/2025/08/29/GRIT-roadmap.pdf (indicating ERCOT plans to implement topology optimization in the first quarter of 2027)

We also recommend that the Commission require all transmission owners to report on asset replacement and maintenance investments to establish a baseline to understand how much of transmission investment is going into this category compared to capacity expansion. As so much grid expansion took place in the 1930s-1980s, the transmission system is naturally in an asset replacement cycle. However, asset replacement decisions must be made with consideration of load growth, changing generation sources, and other dynamics. In many cases, it will be cost effective to use ATTs in addition to or in place of traditional infrastructure.

To understand how much investment is being made without increasing grid capacity, and identify leading transmission owners who are investing strategically to increase capacity, the Commission should require all transmission owners to report on grid investment compared to grid expansion, i.e., how much money they put into system maintenance and replacement, compared to system expansion. A metric could be a ratio of additional transfer capacity added or enabled (potentially scaled by the length of the pathways) per million dollars invested in grid infrastructure. Transmission owners should calculate this metric every year based on past investment and assess whether investment efficiency could be increased through adoption of new technologies.

This illustrative chart shows how a metric of "New Grid Capacity Added / Million Dollars Spent on the Grid" could show the value of grid innovation to unlock more capacity:

Type of upgrade	Time	Additional average capacity (MW) ÷	Cost ($M) =	MW-miles per $M	Note
Asset replacement - reconductor or rebuild	1-2 years	0	5.28	0	1:1 asset replacement may not be the most prudent choice given the pace of load growth.
Rebuild with ACSS	2-3 years	257	22	129	While enhanced-steel cores can increase capacity, they usually require rebuilding towers to fully unlock their capacity, which increases costs.
New line	5+ years	372	23.1	177	New lines add significantly more capacity, but take much more time to come into service, and do not always obviate the need for asset replacement.
New line plus DLR	5+ years	409.2	23.65	190	Building a new line and adding DLR offers more additional capacity than the line alone, though the cost and time-to-service remain high.
Advanced Reconductor with HPCs	1-2 years	243	13.2	203	High Performance Conductors often avoid the need to rebuild towers, so can be in service quickly and at relatively low cost.
APFC		185	14	378	This is an approximation based on a project in New York.[41] APFC deployment value will vary based on grid topology.

[41] SmartWires, *SmartWires Collaborates with Central Hudson Gas & Electric* (Sept. 2024), https://www.smartwires.com/2024/09/18/smart-wires-collaborates-with-central-hudson-gas-amp-electric-to-provide-185-mw-extra-capacity-for-renewable-energy-in-new-york/

Adding DLR to original line	<1 year	37.2	0.55	744	GETs offer the fastest and cheapest addition of new capacity to the grid, so, on a capacity/cost basis, score the highest.

Understanding how transmission owner investments are adding to the grid or maintaining the status quo will help the Commission determine what additional policy is needed to help meet America's energy needs affordably and reliably.

V. Policy lever: encourage operational planning to include efforts to improve grid efficiency

Transmission owners typically do not focus their efforts on improving grid efficiency, instead dedicating internal resources to efficiently planning new investments to meet growing demand and to maintaining existing assets to ensure continued reliability. As a result, very few, if any, transmission owners have employees in engineering or operations roles where one of their primary job responsibilities is to ensure efficient grid operation. The WATT Coalition and its members have engaged in discussions with transmission owners and RTOs around this topic and those conversations revealed that grid operators and operation planning teams see grid reliability as their primary, if not sole, responsibility and do not spend staff resources on grid efficiency, with a few exceptions.[42] On a daily basis (sometimes even hourly), these engineers study and

[42] NewGrid, TRANSMISSION TOPOLOGY OPTIMIZATION: A SOFTWARE GRID-ENHANCING TECHNOLOGY (Jun. 2025), https://www.iso-ne.com/static-assets/documents/100024/2025_06_18_gets_newgrid_materials.pdf; MISO, Process to Support Congestion Cost Reconfigurations in the MISO footprint (Oct. 2024), https://cdn.misoenergy.org/MISO%20Congestion%20Cost%20Reconfiguration%20Process629409.pdf;

implement grid reconfigurations or generation dispatch options to ensure reliable grid operations. In many cases, these interventions may actually create sub-optimal economic dispatch in the interest of preserving reliability margins. However, with GETs, operations engineers and operations planners could find opportunities to both improve economic dispatch and improve reliability. Those processes, however, are not in place.

We urge the Commission to act, through a policy statement or order directing reports, to encourage transmission owners and RTOs to develop processes focused on improving grid efficiency.[43] These processes could unlock latent grid capacity in the timescales needed to win the AI race–the next two to three years—as well as to address the growing affordability crisis. Improvements to operational planning processes likely have the biggest opportunity. RTOs and transmission owners should use fast-to-deploy technologies like GETs to increase capacity while new large-scale infrastructure is built on longer time horizons.

An incremental approach could be instituting new processes at the monthly, quarterly and yearly timescales. This would allow operators to get comfortable with new technologies before implementing them on the daily and hourly timescales.

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[43] Policies like SPP's recently submitted tariff revisions are one example of constructive improvements in RTO operational practices. *See* SPP, *Docket No. ER26-2592-000 Submission of Tariff Revisions to Implement Economic Topology Optimization* (May 2026), https://spp.org/documents/76758/20260521_revisions%20to%20implement%20economic%20topology%20optimizaton_er26-2592-000.pdf

For instance, FERC could require reporting on and encourage development of processes similar to:

- **Monthly operational grid efficiency and reliability analysis**
    - **Outage mitigation:** Quantify projected impacts of planned/scheduled transmission and generation outages (such as for maintenance and construction) and develop outage mitigation practices to reduce economic and reliability impacts. These practices could include analyzing operational and technological remedies (including grid reconfigurations identified with transmission topology optimization software) and business process improvements that could cost-effectively improve economic dispatch and maintain or improve grid reliability.
    - **Congestion intervention:** Where lowest-cost generation has not been deliverable for more than 10 hours in the last month, transmission owners could evaluate and implement grid reconfigurations that maintain reliability and improve deliverability of resources.
- **Quarterly grid efficiency and reliability**
    - **Deployment of DLR or other low-cost targeted upgrades:** Where lowest-cost generation has not been deliverable for more than 20 hours in the last three months due to thermal conductor limits, transmission owners could quantify the benefits that would have been accrued from DLR over the past year on those lines based on historical weather data and calculated

ratings increases. If DLR had been cost-effective and the cause of the constraint had not been addressed by other interventions, DLR should have been deployed to address future congestion. Transmission owners should be prepared to deploy operational DLR on lines within three months, and RTOs should be prepared to integrate new DLR deployments on that timeline. In addition, utilities should be prepared to deploy transmission line coating technologies or tower raising if those interventions would be cost-effective, or other targeted low-cost upgrades.

    - **Outage mitigation:** Where constraints from planned outages are not fully addressed in the monthly grid efficiency analysis, an additional study to determine if DLR would cost-effectively alleviate grid congestion should be common practice, and if the benefit-cost ratio is greater than 1.0 during the expected duration of the outage, DLR should be deployed.

- **Yearly comprehensive efficiency and headroom study**
    - Every year, transmission owners should perform comprehensive study of economic and reliability benefits and grid headroom enabled by ATTs looking out three years into the future. Many states have already required similar practices, including Ohio and California.[44]
    - The benefits must include:

[44] *See* Ohio HB15 from 2025, https://www.legislature.ohio.gov/legislation/136/hb15, and California SB1006 from 2024, https://leginfo.legislature.ca.gov/faces/billVersionsCompareClient.xhtml?bill_id=202320240SB1006.

        - **Reducing energy costs:** transmission owners should evaluate operational and technological interventions and business process improvements to expand capacity and lower the cost of energy.
        - **Expediting grid interconnection capacity for load and generation:** ATTs should be studied for their potential to unlock additional headroom on the grid, which could shorten timelines and reduce costs for generator or load interconnection.
    - ATTs should be studied for these benefits in order from lowest cost to highest cost interventions.
    - The studies should consider planning and construction timelines for new transmission assets, and risks and impacts of potential delays, and evaluate ATTs to cost-effectively increase grid capacity and mitigate constraints while future transmission projects are planned and built.
    - The studies should consider transmission and generation assets that are expected to come into service in the next three years and evaluate whether ATTs could increase the economic or reliability value of those projects by increasing transfer capacity on surrounding grid assets or redirecting power flows.
    - The model should analyze the full suite of ATTs, as well as any required by state law, or additional technologies identified by the transmission owners.
    - Decisions to implement ATTs or not based on these studies should be reported on a regional basis.

- Transmission owners must be prepared to use all technologies studied in these processes in their grid operations by updating relevant software and business processes, and, where needed, selecting vendors.

VI. Policy lever: direct reports and invite further comments to develop a record with updates on GETs availability and deployment progress

The Commission's record on ATT deployments and obstacles to deployments is several years old. The record in Docket No. RM-24-6-000 on the Implementation of Dynamic Line Ratings is largely from before 2024. The Commission's wider investigation of GETs in Docket No. AD 19-19-000 is older still. Though the information and record evidence in those dockets could form the basis of a Notice of Proposed Rulemaking and the Commission has received updates from technology providers at the Annual Software Technical Conferences, record, particularly with regard to transmission owners' perspectives is likely incomplete given recent deployments and operational experience.

The transmission owners that have led the country on GETs deployment have the most up-to-date understanding of the technologies and the technical, institutional, and structural barriers to rapid, full-scale deployment. The Commission's leadership in collecting information on these deployments and the approaches that have unlocked consumer benefits will help scale these approaches to other transmission owners. This information could encourage individual transmission owner deployments, accelerate broader systemic policy changes through proposals to FERC under the Federal Power Act

section 205, and form the basis for FERC action under the Federal Power Act section 206. We would expect leading transmission owners like Great River Energy, Georgia Power, Portland General Electric, Pacific Gas and Electric Corporation, National Grid, Entergy, Central Hudson, VELCO and others to have useful experience to share. In addition, all the RTOs have taken incremental steps on ATTs and will have learnings to share. We believe that there is a much longer list of leading transmission owners and more efforts underway at the RTOs than the WATT Coalition is even aware of.

Therefore, we suggest that the Commission direct reports and invite further comments in an open or new docketed proceeding, focusing in particular on transmission owners and RTOs to refresh the Commission's record around the state of ATT adoption. The following questions are a starting point for identifying success stories and lessons learned, and policy changes that may support more effective ATTs adoption.

1. Which transmission technologies is your company using or planning to use in grid planning, operational planning, and operations, which could provide speed to power for large loads while protecting ratepayers from potential cost increases?
    a. Ambient Adjusted Line Ratings
    b. Dynamic Line Ratings
    c. Advanced Power Flow Control
    d. Transmission Topology Optimization through Transmission Switching
    e. Advanced Conductors (including ACCC, AECC, and superconductors)
    f. Tower Raising

g. Emissivity Coatings

h. Other

2. How has your company prepared to integrate each of the above-listed technologies into grid planning, operational planning, and operations?
3. How has your company investigated or deployed the above-listed technologies in generator or load interconnection?
4. Which teams in your company are engaged on the integration of each of the above technologies into grid planning, operational planning, and operations at your company?
5. Therefore, we encourage FERC to issue a policy statement with further guidance on what kinds of incentives for ATTs projects might be just and reasonable under the existing requirements of Order No. 679.
6. What barriers to implementation exist for the above technologies in operational planning and operations at your company?
7. What process changes would facilitate integration of the above technologies into operational planning and operations?
8. Do your company's operational planning processes and practices include:
    a. Reducing impacts of outages through grid reconfiguration/transmission switching/topology optimization;
    b. Reducing impacts of thermal constraints (including temporary, outage-driven constraints) with Dynamic Line Ratings;

c. Deploying Advanced Power Flow Control or High Performance Conductors to increase grid capacity for near-term needs?

9. What, if any, North American Electric Reliability Corporation standards should be updated to facilitate integration of the above technologies into grid planning, operational planning, and operations?
10. Do you have any internal company thresholds for considering a GETs intervention in the operations or operations planning workflows that trigger a review of GETs interventions, e.g. a certain level of congestion on a thermally-limited line or a scale of planned outage?

VII. Policy lever: encourage regional forums and host FERC technical conference to share utility progress

In addition to submitting written comments to the Commission as suggested in Section IV, the transmission owners and RTOs could support faster progress on ATTs deployment through regional knowledge-sharing and collaboration.

Thus we urge the Commission to encourage all RTOs to host forums to investigate the opportunity for ATTs and other solutions to increase capacity and improve affordability and how to integrate ATTs into all aspects of an RTO's processes. All transmission owners should participate in one of these forums, and those not in RTO regions should likewise coordinate an alternative forum for their regions.

Following these forums, in 2027, the Commission could convene a technical conference to hear from representatives from each region about progress in modernizing grid operations and operational planning to include more commercially available solutions. The agenda should cover regional strategies for effective load integration and efficient grid interconnection and operations, based on new processes and the use of technologies.

Topics that these forums could cover include:

- How each ATT affects the different activities and areas of responsibility within RTOs and utilities, based on operational deployments in the regions, and examples of how control room operators use the technologies in specific circumstances.
- How modelers and engineers can study and design ATTs deployments across standard utility software for production costs, capacity expansion and power flow modeling.
- Process and system improvements needed to unlock the benefits of ATTs for reliability and resilience, interconnection of load and generation, and market efficiency.

VIII. Policy lever: encourage new ATTs-specific transmission incentives

Section 219 of the Federal Power Act authorizes FERC to establish appropriate incentives to encourage transmission development.[45] This includes encouraging ATTs.[46] FERC Order No. 679, issued pursuant to that statutory authority, allows transmission owners to propose packages of incentives for transmission investment that are "tailored to address the demonstrable risks or challenges faced by the applicant," and create just and reasonable rates.[47] But the Order stops short of clarifying the incentives that could be proposed to encourage accelerated and widespread deployment of ATTs.

Order No. 679 does enumerate options for incentive enhancements, including return on equity (ROE) adders to attract investment and ensure cost recovery if the transmission owner has to cancel the project for reasons outside of its control. The Order also leaves the door open for "any other incentives approved by the Commission, pursuant to the requirements of this paragraph, that are determined to be just and reasonable and not unduly discriminatory or preferential."[48] In 2019, FERC reported that it had acted on 109 incentive applications for more than $80 billion in anticipated construction costs, but we have not heard of any proposals for ATTs to receive incentive treatment.

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[45] 16 U.S.C. 824s(b)(3): Requiring FERC to establish a rule with "incentive-based (including performance based" incentives for transmission, and the rule shall "encourage deployment of transmission technologies and other measures to increase the capacity and efficiency of existing transmission facilities and improve the operation of the facilities."

[46] *Initial Comments of the WATT Coalition*, Inquiry Regarding the Commission's Electric Transmission Incentives Policy, Docket No. PL19-3-000 (filed June 26, 2019).

[47] *Promoting Transmission Investment Through Pricing Reform*, Order No. 679-A, FERC Stats. & Regs. ¶ 31,236 at P 27 (2006)

[48] 18 C.F.R. § 35.35(d)(6) (2006).

Therefore, we encourage FERC to issue a policy statement with further guidance on what kinds of incentives for ATTs projects might be just and reasonable under the existing requirements of Order No. 679.[49]

An additional incentive for large-scale ATTs deployment could be appropriate in order to compensate:

1. Significant staff time validating, designing and implementing an ATTs deployment
2. Uncertainty about the level of realized benefits, based on potential changing market conditions for generation
3. Costs related to software and process updates
4. Costs related to staff upskilling; and/or
5. Need to attract capital to fund the deployment of technologies that the transmission owner has not deployed before, which may be seen as a riskier investment.

ATTs incentives could be structured as:

1. ROE adders
    a. Australia requires transmission owners to prove consumer benefits before approving new transmission projects through its Regulatory Investment

[49] *FERC Opens Inquiry on Improvements to Electric Transmission Incentives Policy*, News Release, Docket No. PL19-3-000 (Mar. 21, 2019), https://www.ferc.gov/news-events/news/ferc-opens-inquiry-improvements-electric-transmission-incentives-policy.

Test for Transmission (RIT-T). The Network Capability Incentive Parameter Action Plan (NCIPAP) adds revenue incentives for operational improvements, directly encouraging ATT adoption.

2. Shared savings incentives
    a. The WATT Coalition has proposed a federal shared savings incentive that allows transmission owners to profit based on a portion of the congestion cost savings from GETs deployments. This directly links utility profits to consumer benefits and operational efficiency.
    b. The UK's RIIO (Revenue = Incentives + Innovation + Outputs) framework pays transmission owners for performance outcomes, not just capital spending. This total expenditure ("totex") approach reduces the bias to build and allows the regulator to set cost expectations and offer shared savings incentives for transmission owners to beat those estimates.

IX. Conclusion

ATTs can deliver meaningful increases in transmission capacity to facilitate the interconnection of large load in less than three years. While recent deployments by PG&E, Georgia Power and others validate this potential, utility adoption remains limited. In addition to the interconnection of large loads, ATTs can reduce congestion, lower costs for consumers, and improve reliability.

We recommend that the Commission require consideration of ATTs in large load interconnection studies, consistent with recommendations from a broad range of

stakeholders. However, limiting ATTs to the interconnection process alone will not unlock their full value. To achieve the ANOPR's goals of enabling faster load interconnection, supporting economic growth, and maintaining affordable and reliable electric service, we also recommend that the Commission encourage and facilitate the broader integration of ATTs into transmission planning, operational planning, and system operations.

Accordingly, the WATT Coalition urges the Commission to: (1) require reporting and studies that quantify the potential for GETs and other ATTs to address congestion and improve transmission investment efficiency; (2) encourage transmission owners and RTOs to establish operational processes focused on increasing grid efficiency and unlocking available capacity; (3) refresh the Commission's record on ATT deployment through targeted information gathering; (4) support regional forums and technical conferences to accelerate knowledge sharing and best practices; and (5) provide guidance on transmission incentives for technologies increase the capacity and efficiency of existing transmission facilities and improve the operation of the facilities. These steps can be taken quickly and show results in the next year. Their follow-on effects will also help with longer-term policy, including potentially finalizing the rule proposed in Docket No. 24-6-000 on the Implementation of Dynamic Line Ratings with a threshold-based requirement for DLR deployments.

Respectfully submitted,



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WATT Coalition

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541-908-5792

X. Appendix A: Ampacimon-Basin DLR Experience, DTECH Midwest 2025

BASIN ELECTRIC POWER COOPERATIVE

A Touchstone Energy® Cooperative

Ampacimon

Ampacimon-Basin DLR Experience

DTECH Midwest 2025

Jeremy Severson (Basin) and Corey Hauser (Ampacimon)

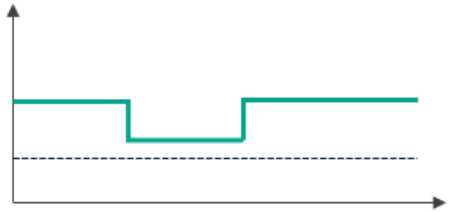
Various Types of Line Ratings

Considers worst case: least cooling



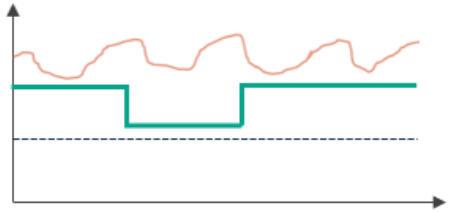
Static Line Rating (SLR)

Considers seasonal worst case



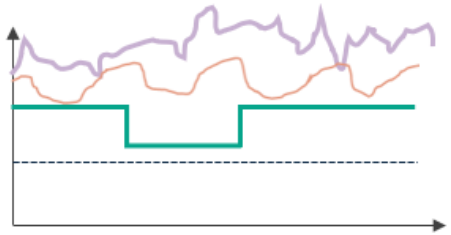
Seasonally Adjusted Rating (SAR)

Considers air temperature cooling



Ambient Adjusted Rating (AAR)
5-10% gains

+ Considers wind cooling

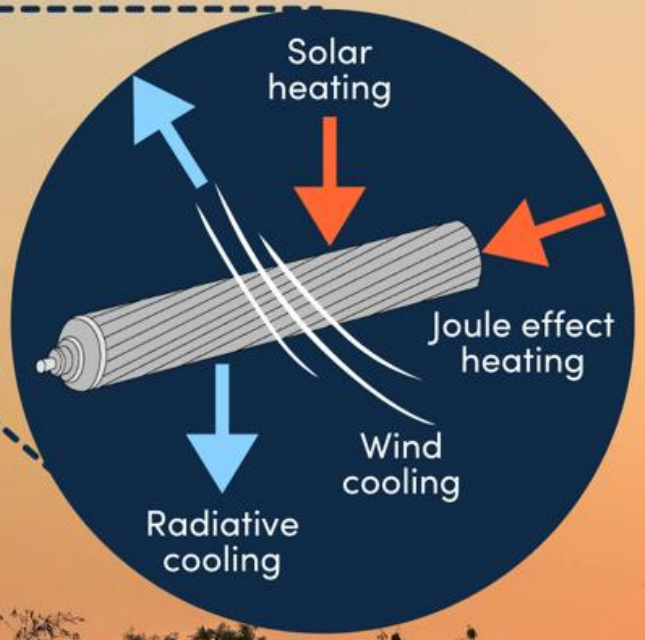


Dynamic Line Rating (DLR)
10-50% gains

Importance of Wind on Ratings

Operating Conditions	Change in Conditions	Impact on Capacity
Ambient temperature	2 °C decrease 10 °C decrease	+ 2% + 11%
Solar radiation	Cloud shadowing Total eclipse	+/- a few percent + 18%
Wind	3 ft./s increase, 45° angle 3 ft./s increase, 90° angle	+ 35% + 44%

Source: Dynamic Line Rating Systems for Transmission Lines Topical Report, US Dept of Energy, April 25, 2014

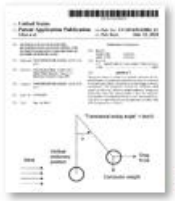


Ampacimon ADR Sense Sensor

Direct Measurement | Wind Speed | Easy Installation | High Accuracy | High Data Availability



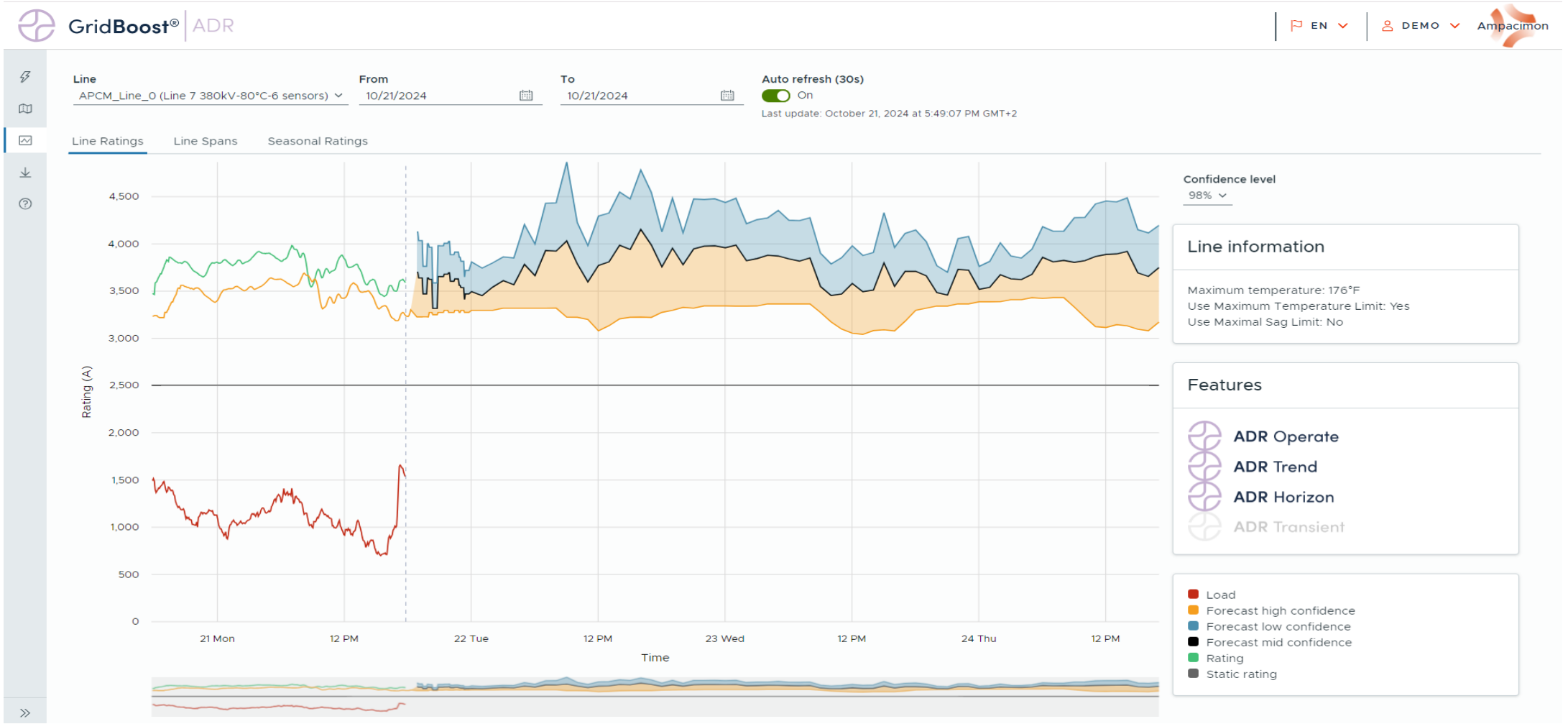
- Vibration-based metrology requires no calibration
- Direct Measurement of Perpendicular Wind Speed
- Machine learning for highest accuracy ratings forecasts
- High Accuracy Sag and Clearance Measurements at all conductor elevations
- Cellular or Satellite Communications
- Measurement of Mean Conductor Temperature
- EMS Integration Through DNP3
- Self-Powered on 10 kV – 700 kV lines
- Live line installation in less than 5 minutes
- Embedded Tension Sensor for Ice Detection and Conductor Health



Protected by 3 patents

ADR SENSE

Ampacimon DLR User Interface



PPL Use Case

	Reconductor	Rebuild	Dynamic Line Rating
			
Time to Implement	2 – 3 Years	3 – 5 Years	~1 Year
Downtime	Extended Outages	Extended Outages	No Outages
Cost	$0.5 M per mile	$2 - 3 M per mile	<$50k per mile*
Est Capacity Benefit	+ 34%	+ 106%	+ 10 – 30%

*< $1 M Total cost on 20-mile line & longer lines are cheaper

Basin Electric Snapshot

139 Members in 9 States

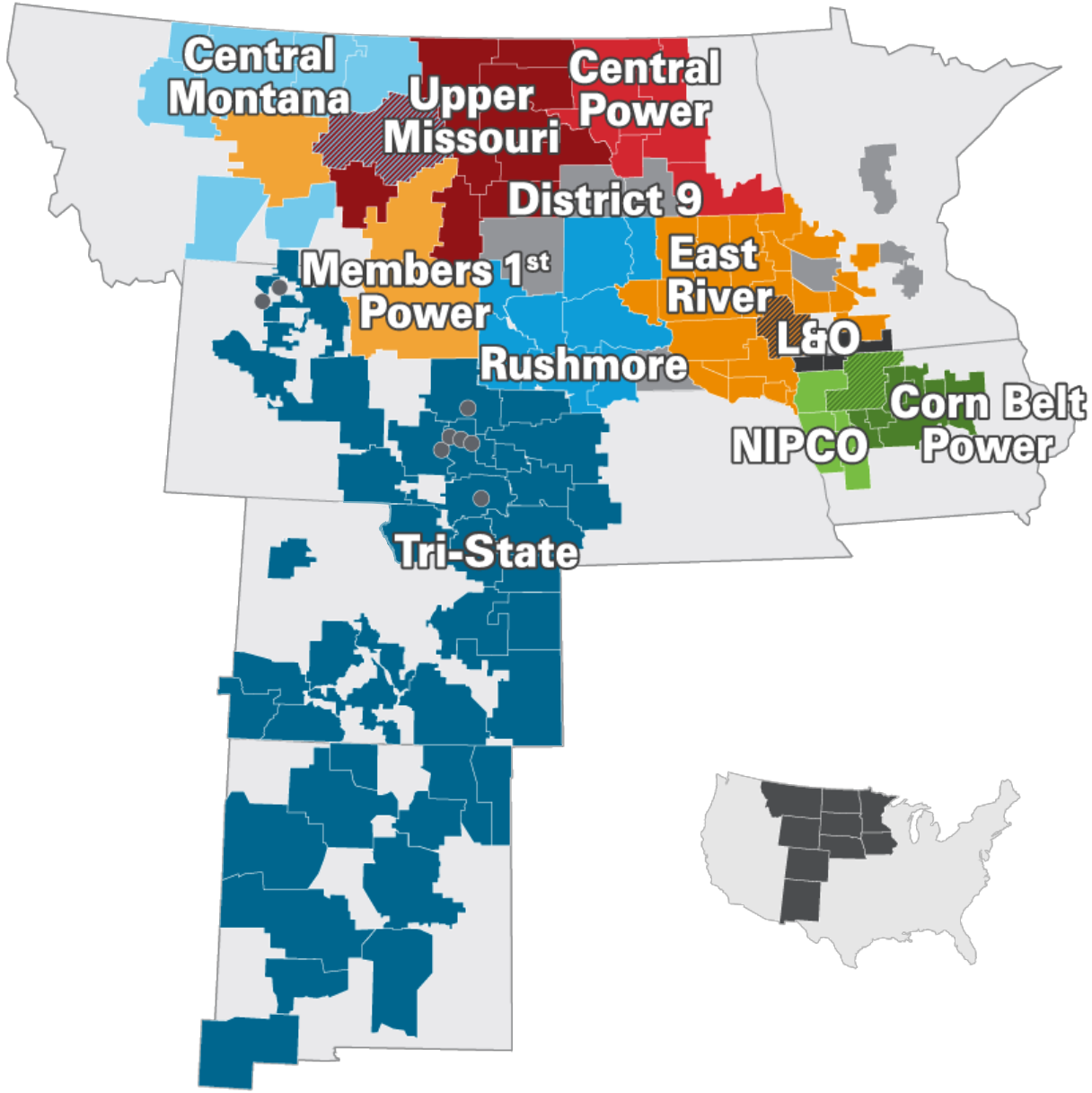
3 million member-owners

Nearly 8,500 megawatts of winter capacity

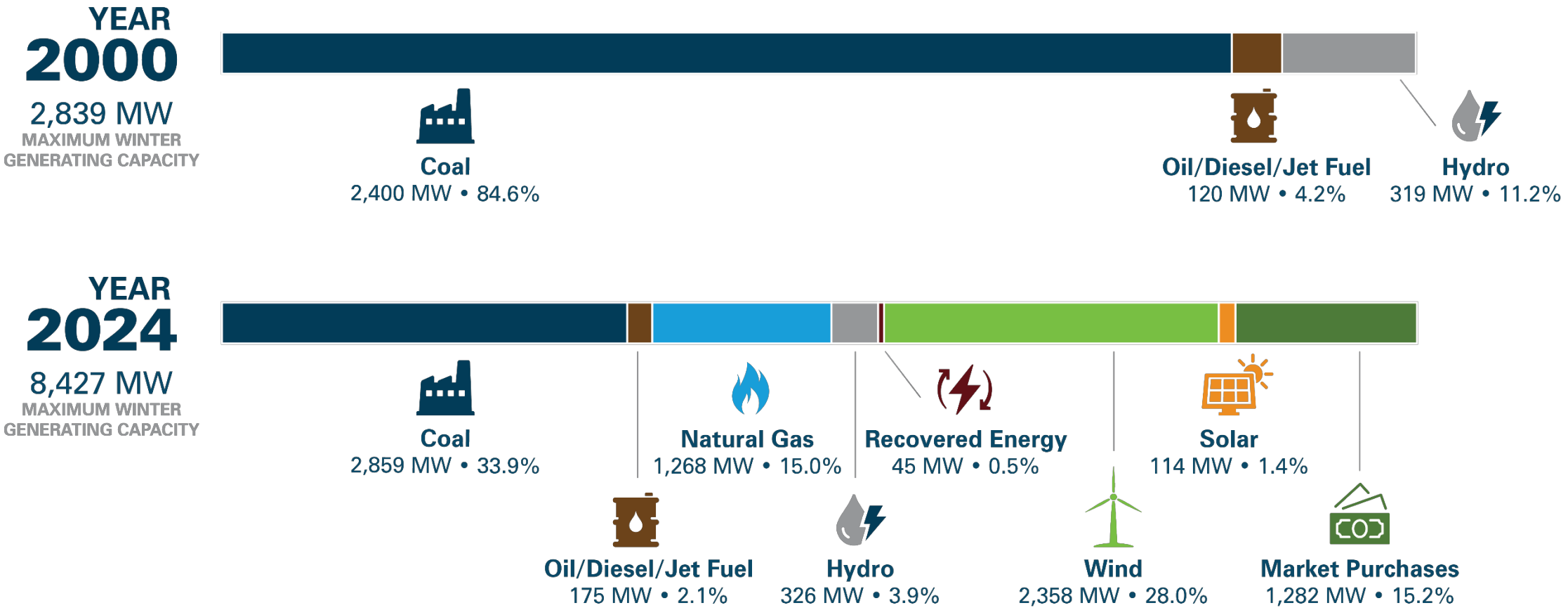
Largest G&T:
- Total MWh sales
- Member sales
- Total Operating Revenue
- Geographic territory served

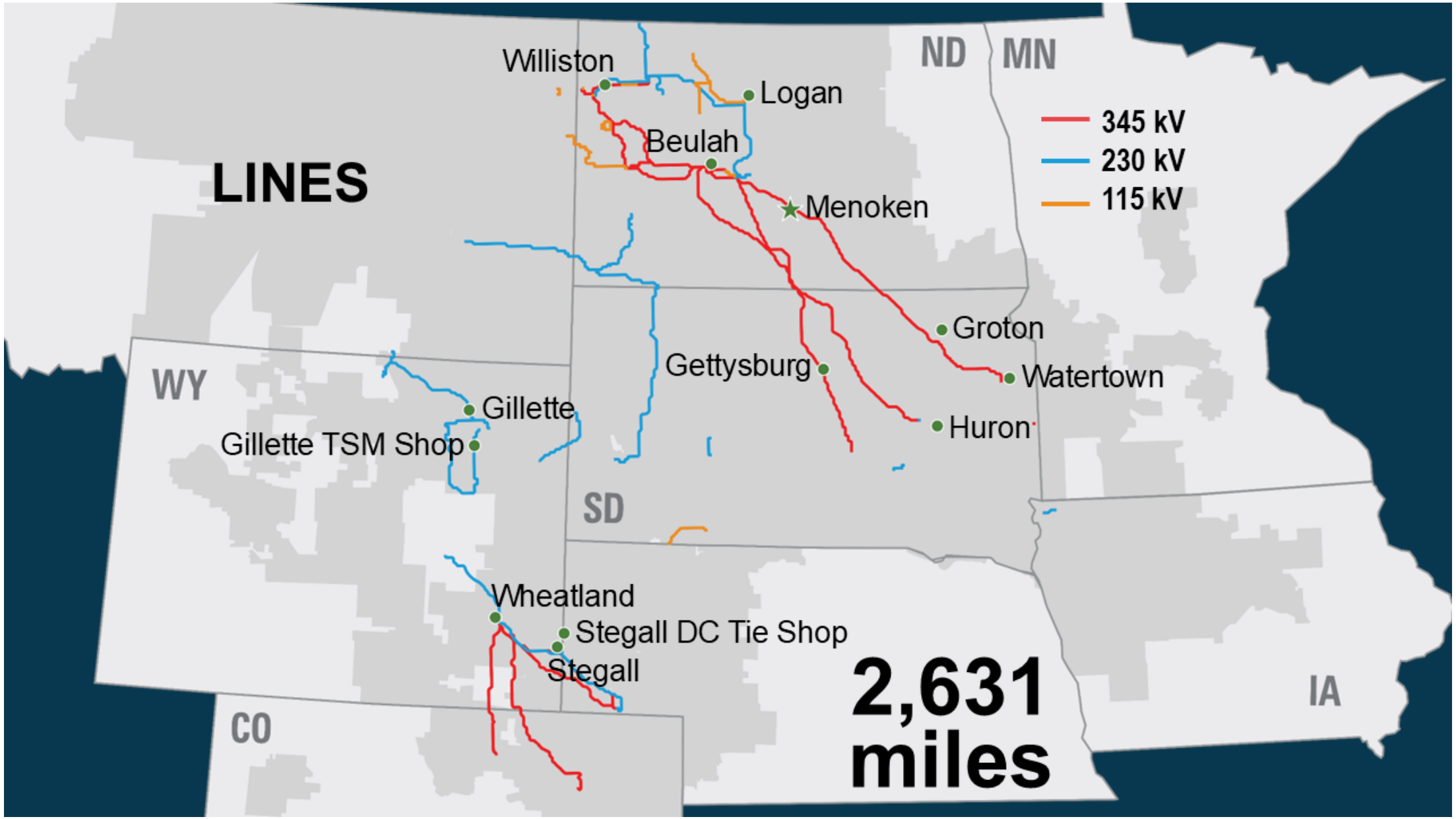
2nd largest G&T by assets

January 2025 Billing Peak Demand Record – 5,150 MW

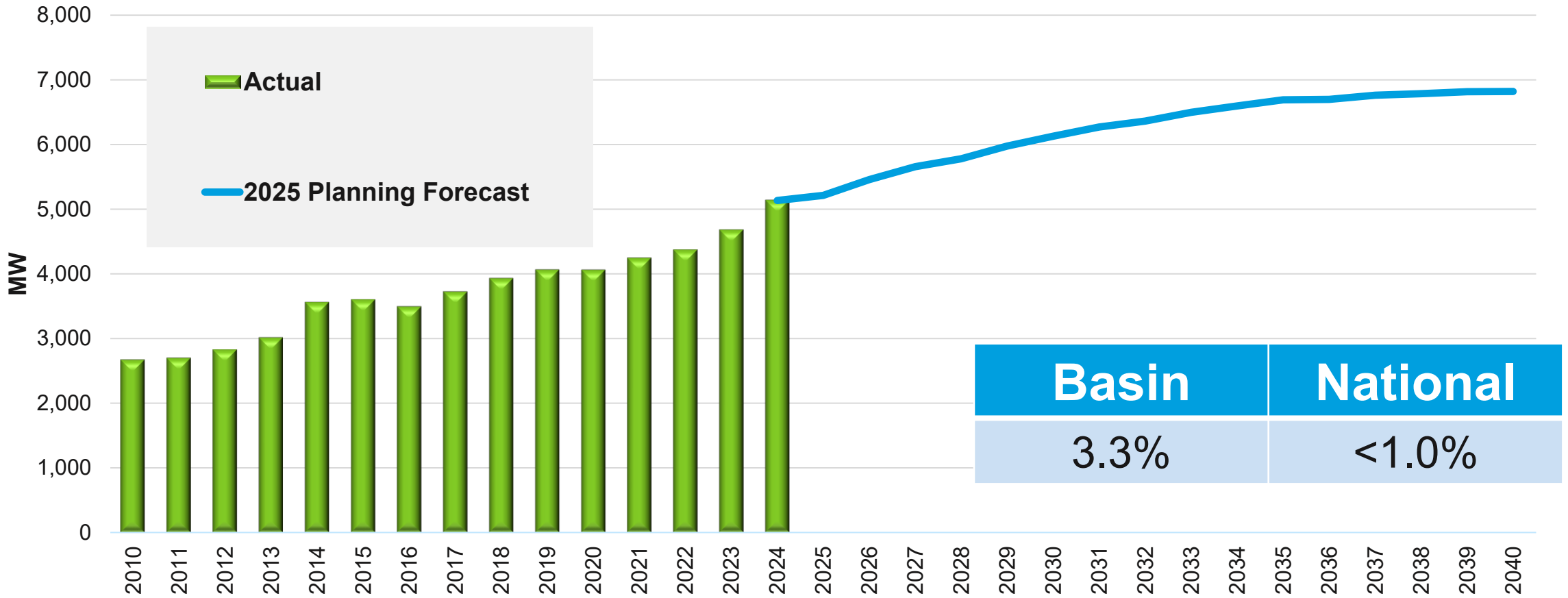


Annual Generating Capacity



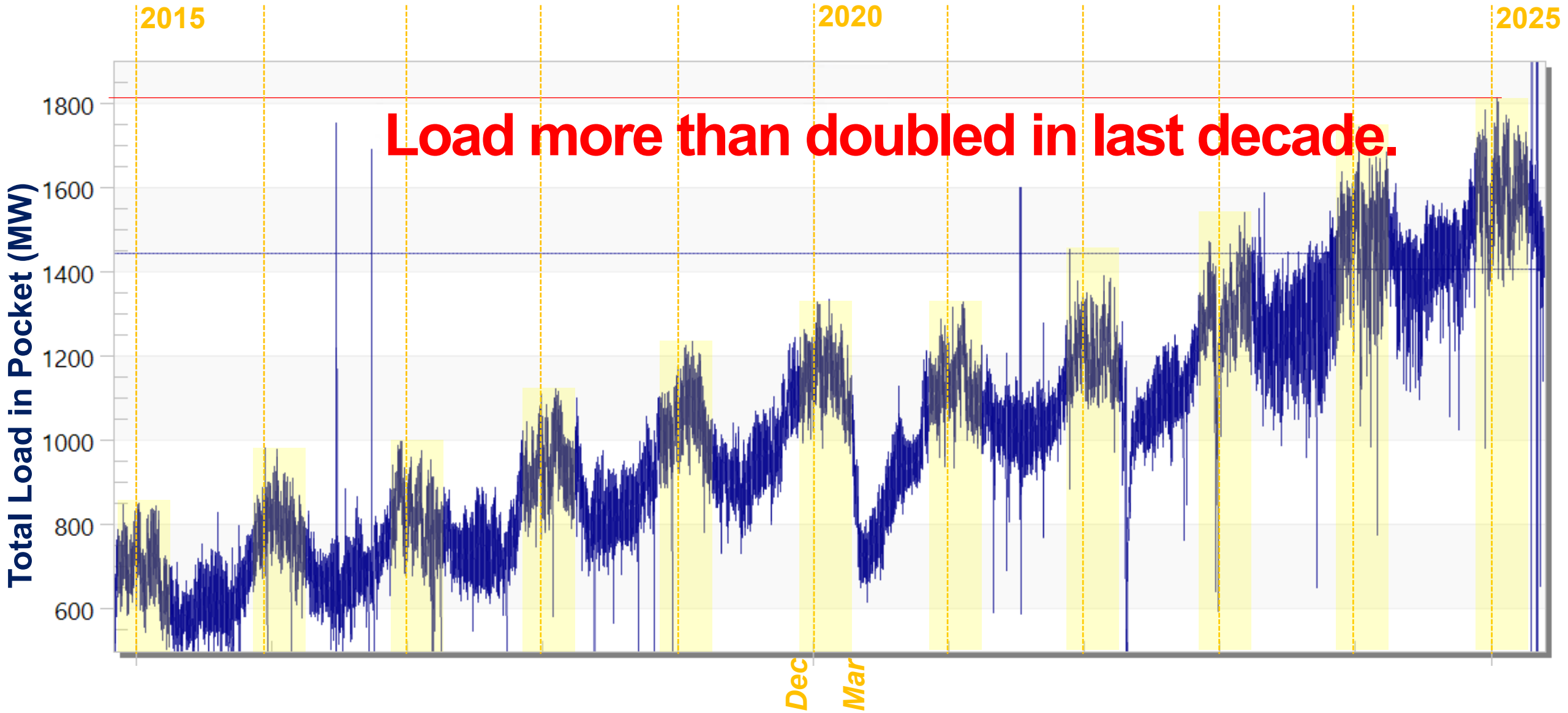

LINES
345 kV
230 kV
115 kV
Williston
Logan
Beulah
Menoken
Groton
Watertown
Gettysburg
Huron
Gillette
Gillette TSM Shop
Wheatland
Stegall DC Tie Shop
Stegall
2,631
miles
ND
MN
WY
SD
CO
IA

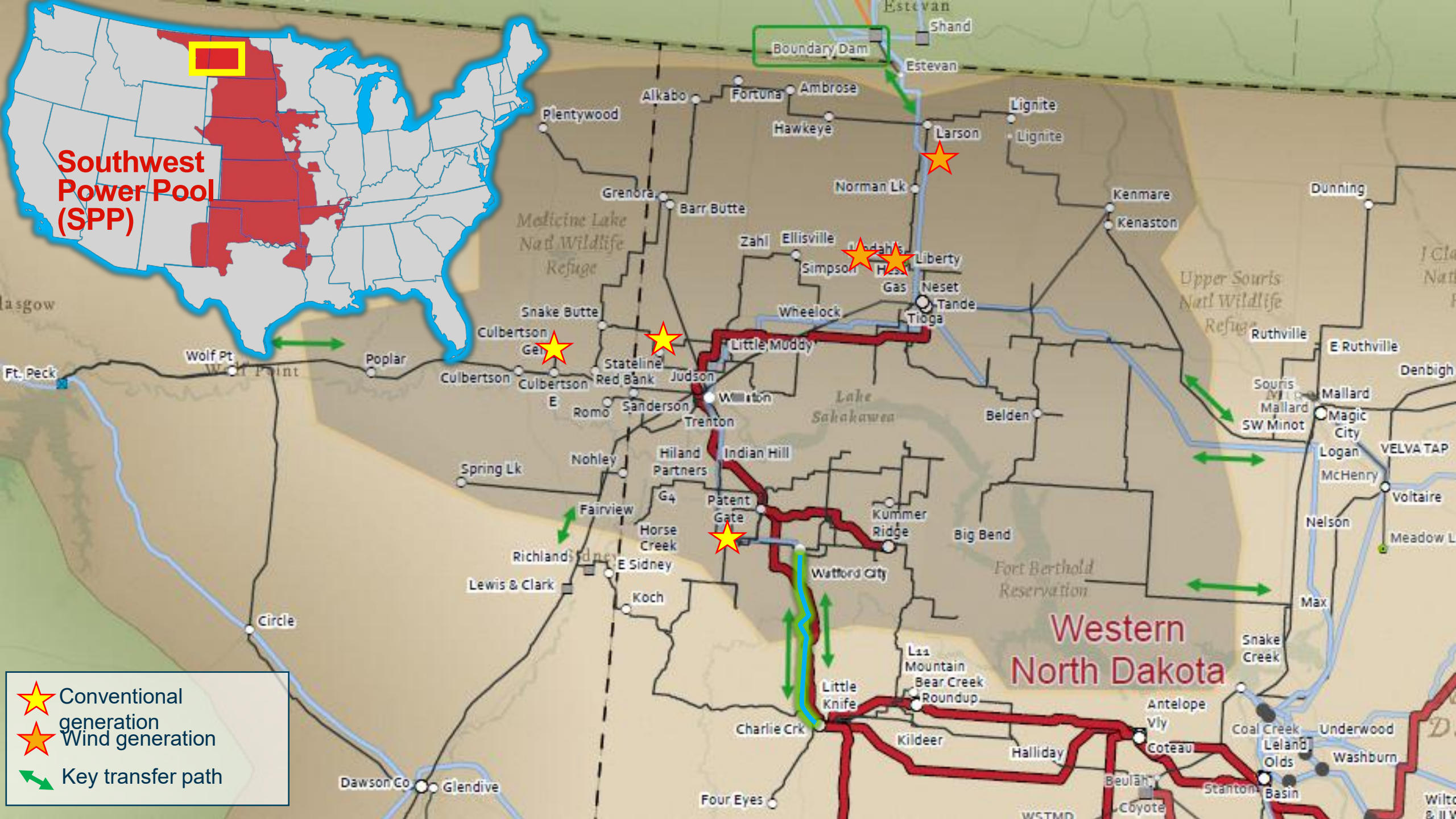
2025 Basin Electric Load Forecast



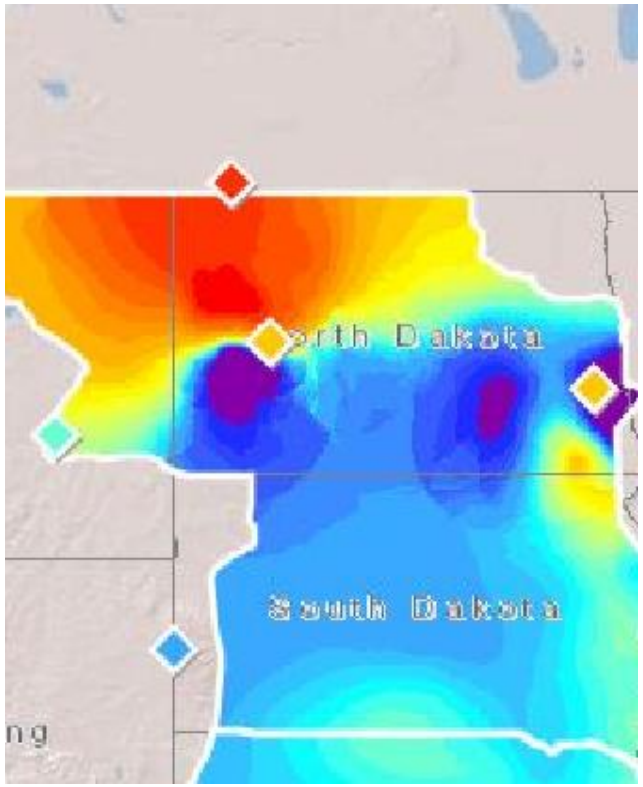
Basin	National
3.3%	<1.0%

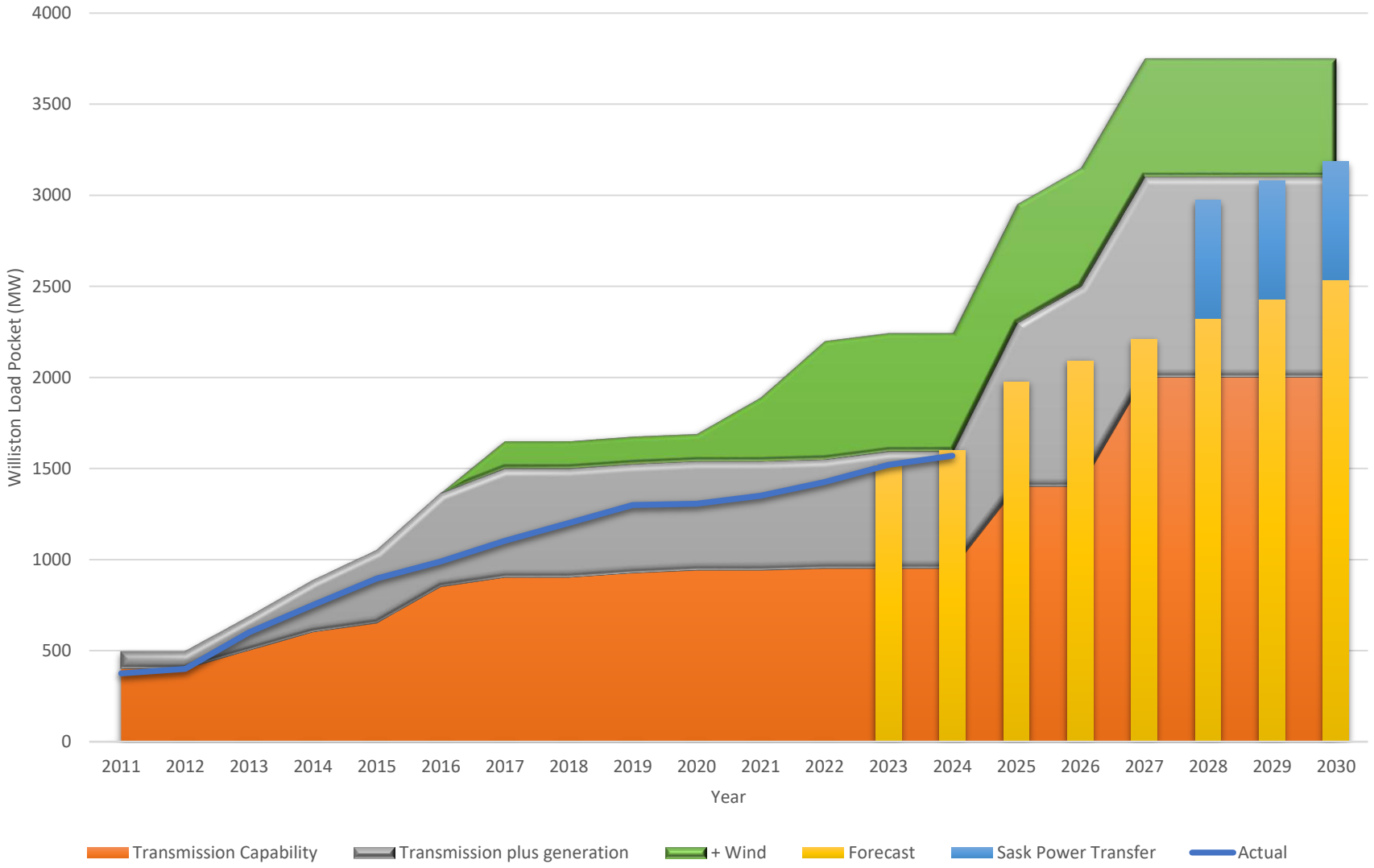
Northwest North Dakota (NWND) Load Area

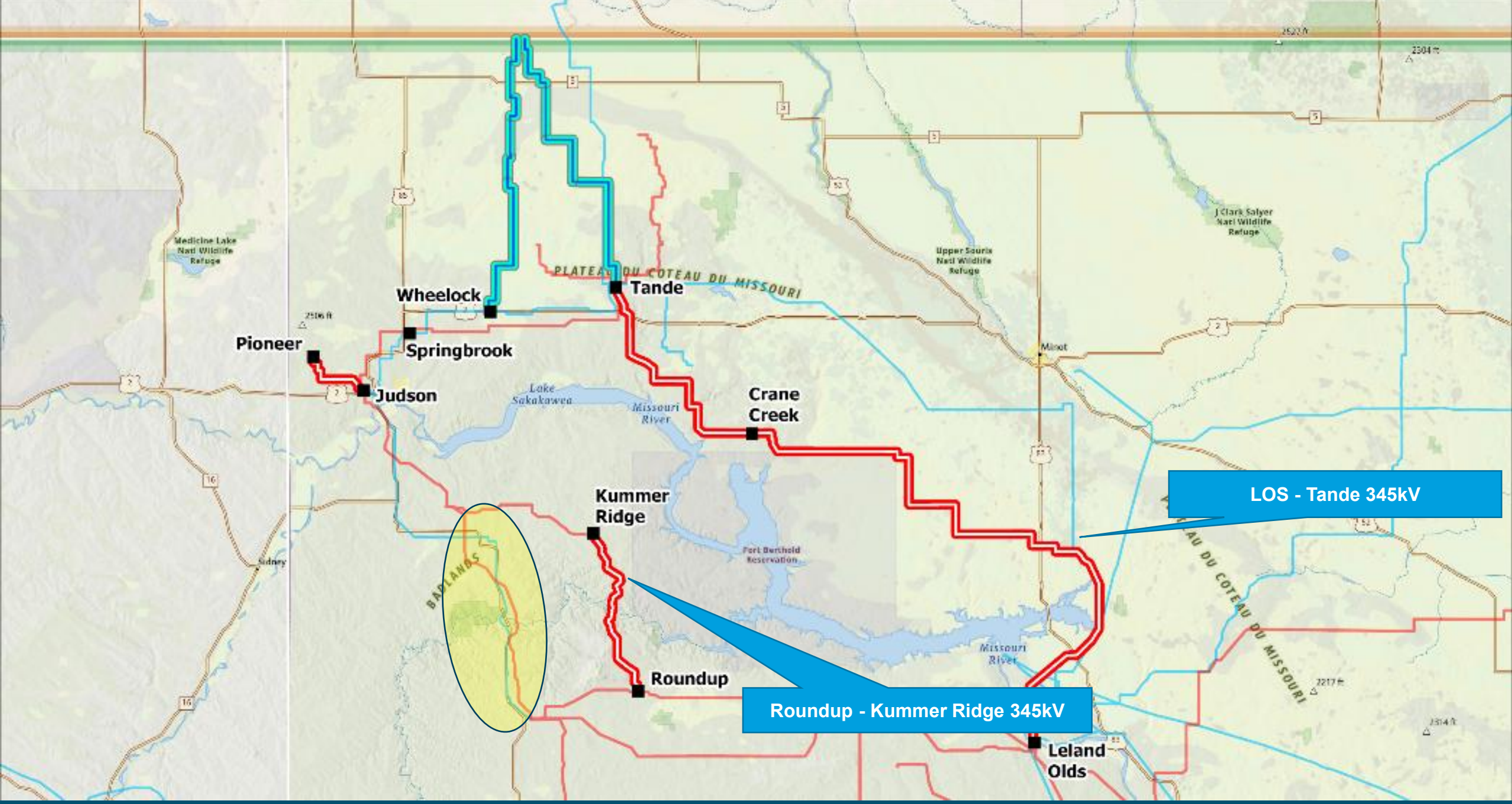



Southwest Power Pool (SPP)
Western North Dakota
Conventional generation
Wind generation
Key transfer path
Boundary Dam
Estevan
Shand
Alkabo
Fortuna
Ambrose
Plentywood
Hawkeye
Lignite
Larson
Norman Lk
Grenora
Barr Butte
Kenmare
Kenaston
Dunning
Medicine Lake Natl Wildlife Refuge
Zahl
Ellisville
Simpson
Liberty
Gas
Neset
Tande
Tioga
Wheelock
Upper Souris Natl Wildlife Refuge
Ruthville
E Ruthville
Denbigh
Snake Butte
Culbertson Gen
Little Muddy
Wolf Pt
Poplar
Ft. Peck
Culbertson
Culbertson E
Stateline
Red Bank
Judson
Romo
Sanderson
Trenton
Lake Sakakawea
Belden
Souris
Mallard
Mallard
SW Minot
Magic City
Logan
VELVA TAP
McHenry
Voltaire
Nelson
Meadow L
Spring Lk
Nohley
Hiland Partners
Indian Hill
G4
Patent Gate
Fairview
Horse Creek
Kummer Ridge
Big Bend
Richland
E Sidney
Watford City
Fort Berthold Reservation
Lewis & Clark
Koch
Max
Circle
Snake Creek
Lit Mountain
Bear Creek
Roundup
Little Knife
Antelope Vly
Coal Creek
Underwood
Leland Olds
Washburn
Charlie Crk
Kildeer
Halliday
Coteau
Beulah
Stanton
Basin
Dawson Co
Glendive
Four Eyes
Coyote
WSTMD

Transmission Limits

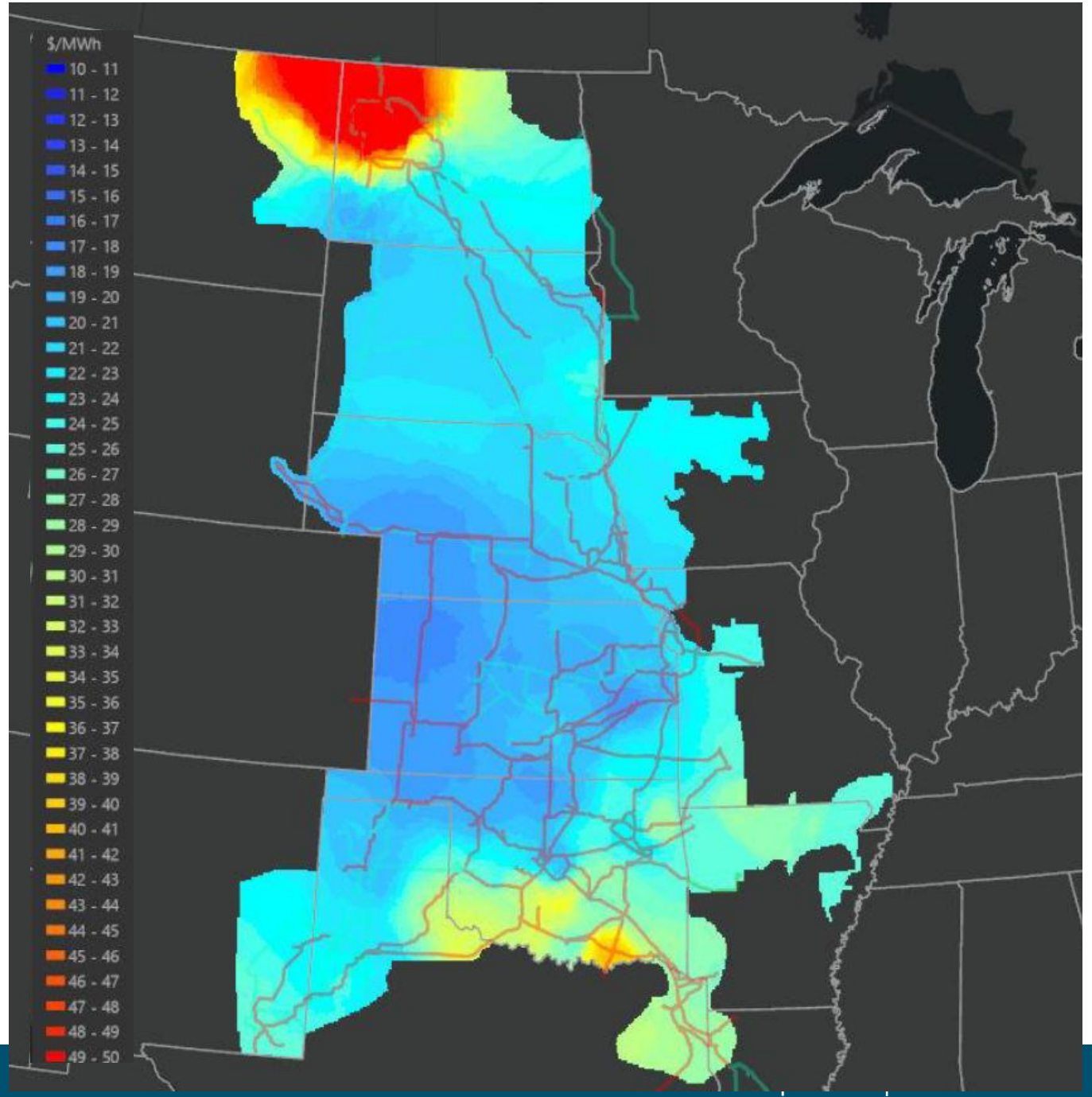





Wheelock
Tande
Pioneer
Springbrook
Judson
Crane Creek
Kummer Ridge
Roundup
Leland Olds
LOS - Tande 345kV
Roundup - Kummer Ridge 345kV
Lake Sakakawea
Missouri River
Fort Berthold Reservation
BADLANDS
PLATEAU DU COTEAU DU MISSOURI
Medicine Lake Natl Wildlife Refuge
Upper Souris Natl Wildlife Refuge
J Clark Salyer Natl Wildlife Refuge
Minot
Sidney

Congestion = Friction

- In 2023, the single most congested transmission line in **all of SPP** was in northwestern North Dakota.
- Accrued $126M of congestion costs between June and October alone.
- Earliest expected relief from new transmission was 18 months away.



Project

- System Need:
  - Thermal rating issue during peak load situations
  - Long term mitigation (new transmission) under construction

- Options:
  - TOP historically been utilizing temperature adjusted operating guides.
    - Inefficient and difficult for day ahead and real time analysis
  - Opportunity for Dynamic Line Rating (DLR) efficiencies

- Solution:
  - Implement DLR by the Summer of 2024

DLR Sensor Installation

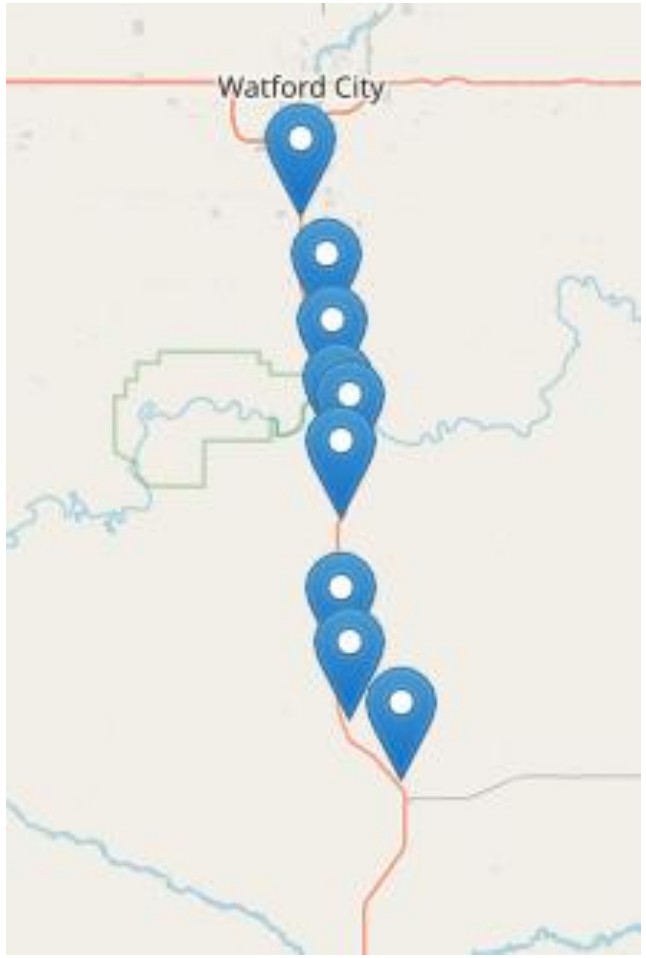


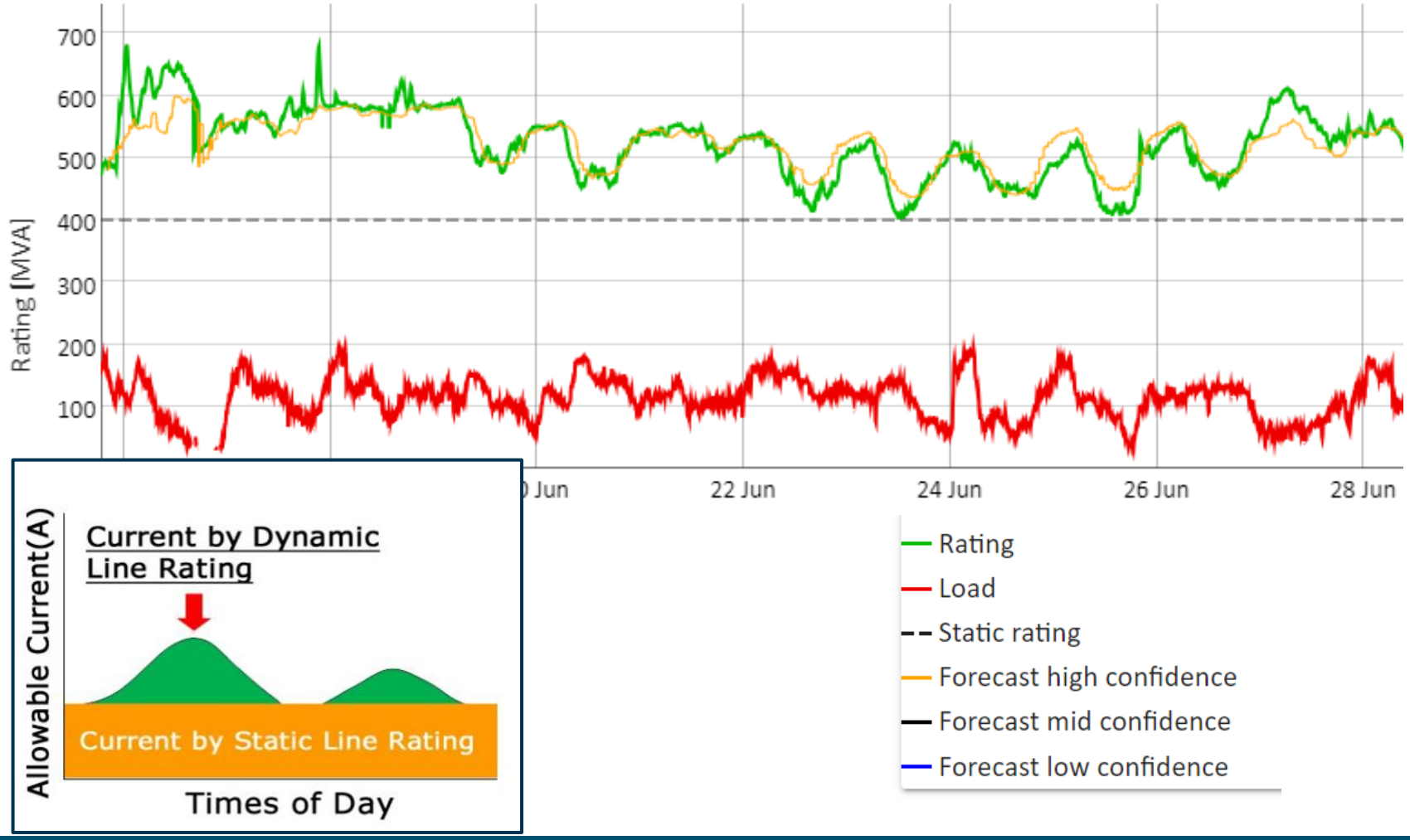






Dynamic Line Rating Impact





Assess Operational Value of DLR

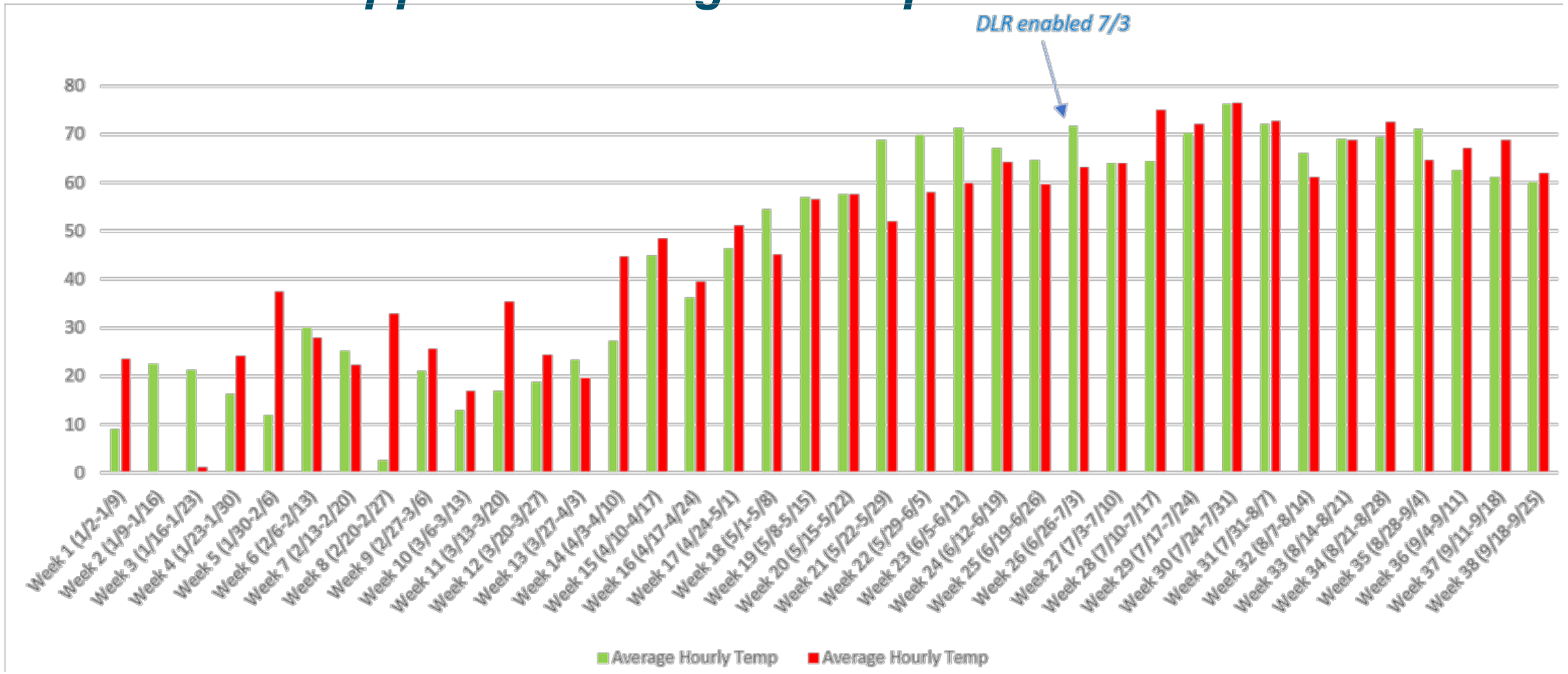
Apples-to-oranges comparison

- DLR was immediate solution to severe market constraints for summer 2024.
- Difficult to make direct comparison of before & after DLR.
- Why?
  - Ambient conditions vary.
  - Load grew.
- Let's review the conditional differences 2023 & 2024.

  → *Focus on summer after DLR enabled (7/3/2024).*

Average air temperatures (weekly basis)

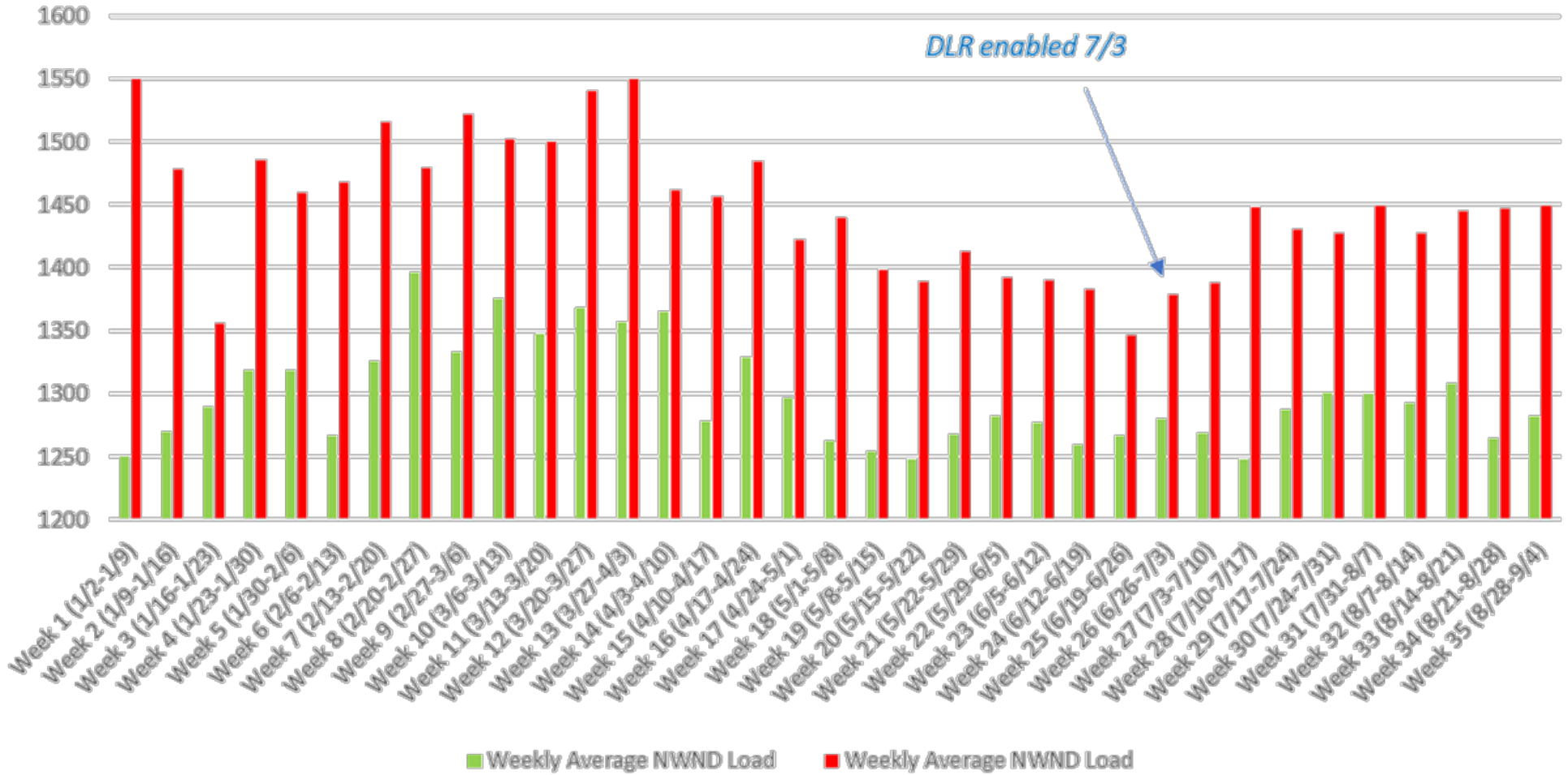
Apples-to-oranges comparison



2023 and 2024 are reasonably comparable.

Average NWND loading (weekly basis)

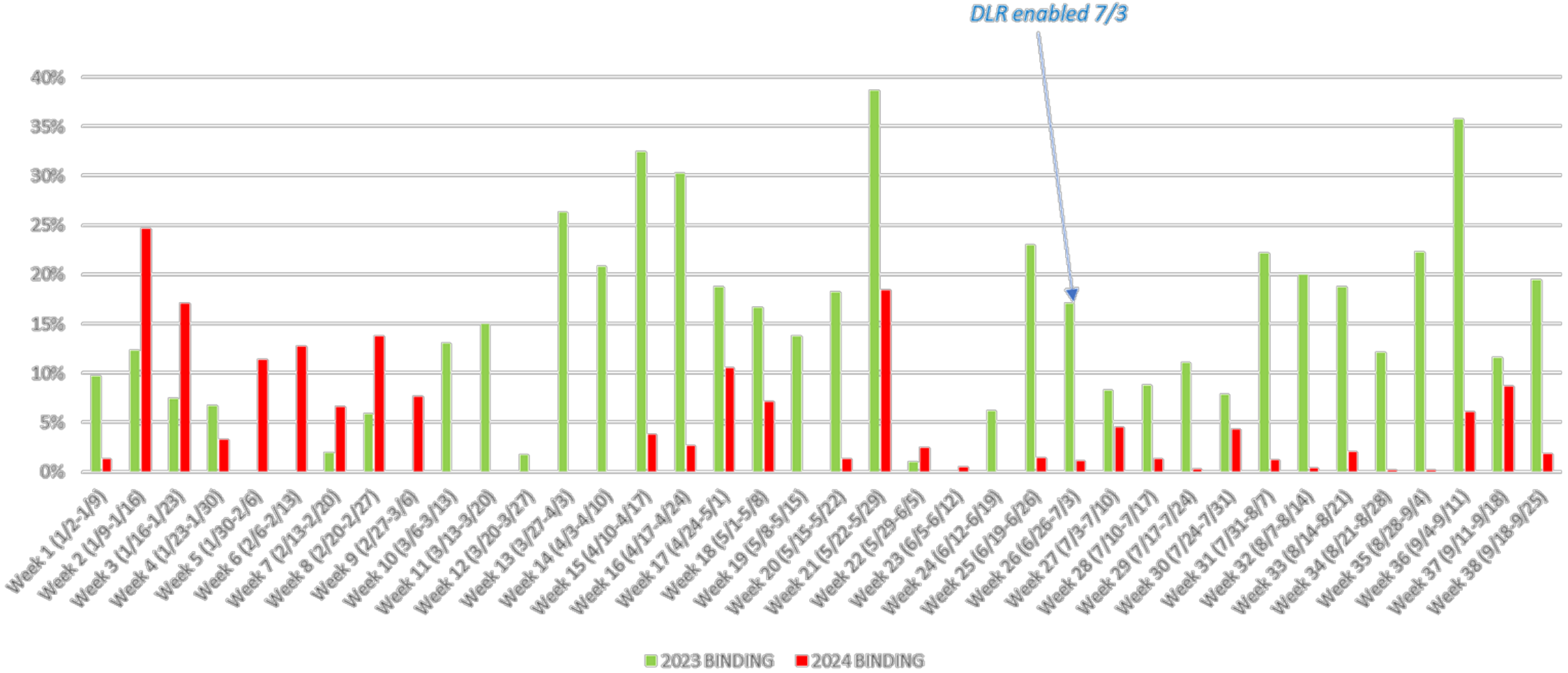
Apples-to-oranges comparison



Significantly more load in 2024.

Real-time market flowgate binding (weekly basis)

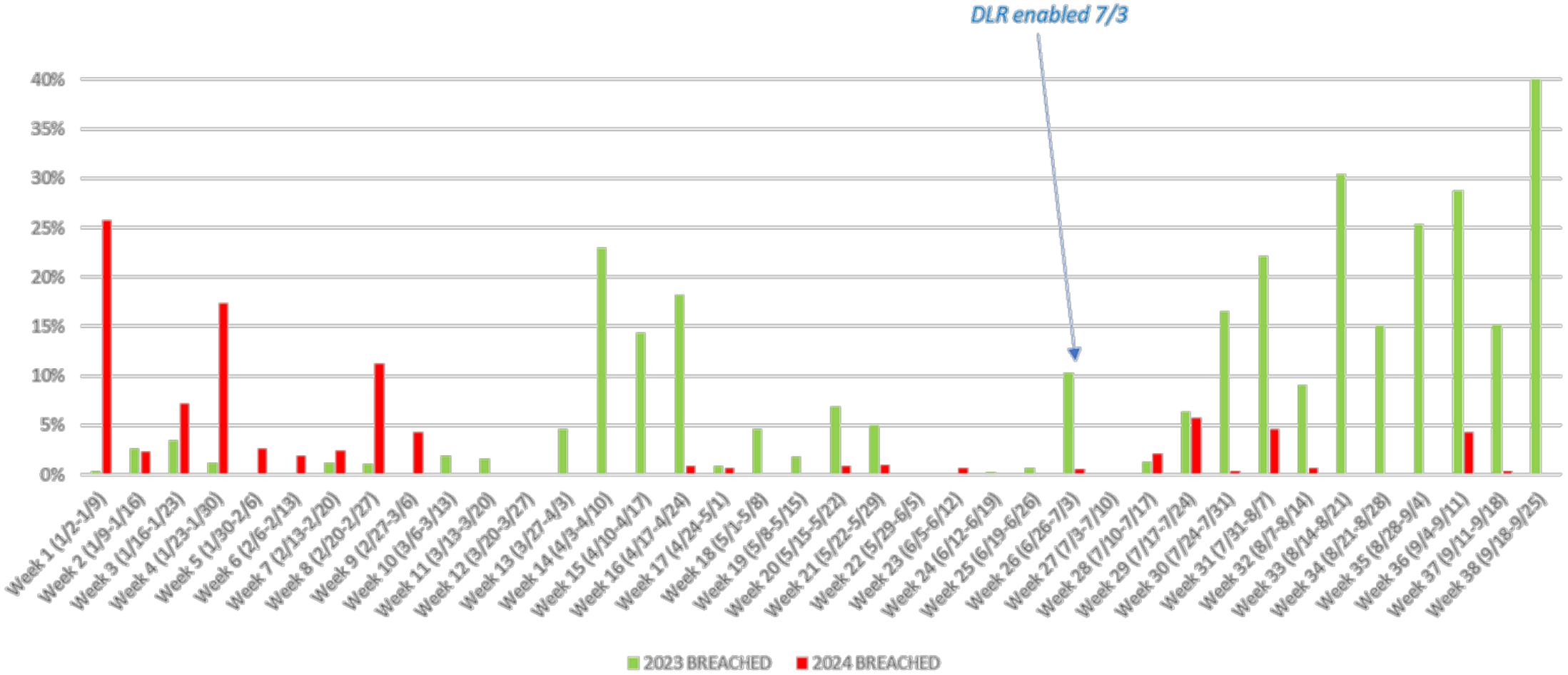
Apples-to-oranges comparison



Dramatically reduced flowgate binding in summer 2024.

Real-time market flowgate breaching (weekly basis)

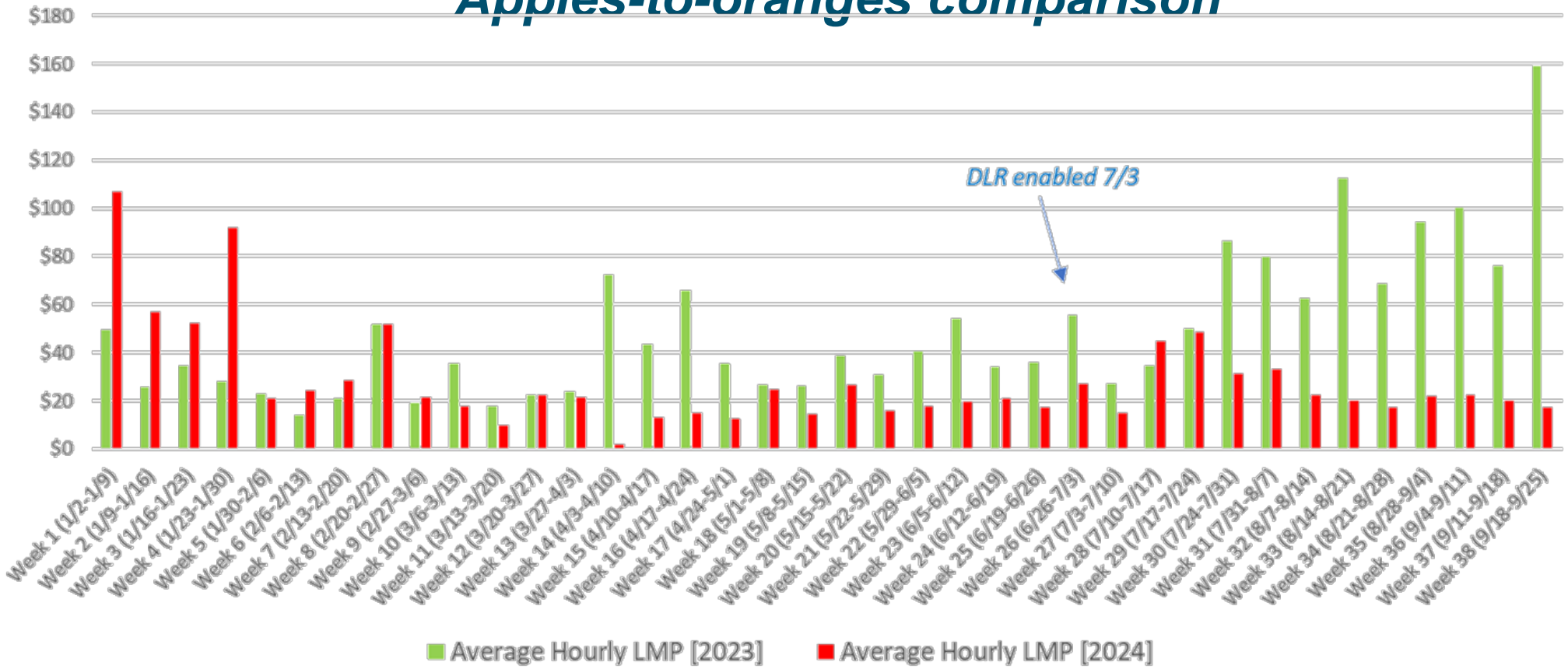
Apples-to-oranges comparison



Virtually eliminated flowgate breaches in summer 2024.

Average Real-time energy prices (5-min LMP; weekly basis)

Apples-to-oranges comparison



Significantly lower delivered energy costs for load customers in summer 2024.

Takeaways

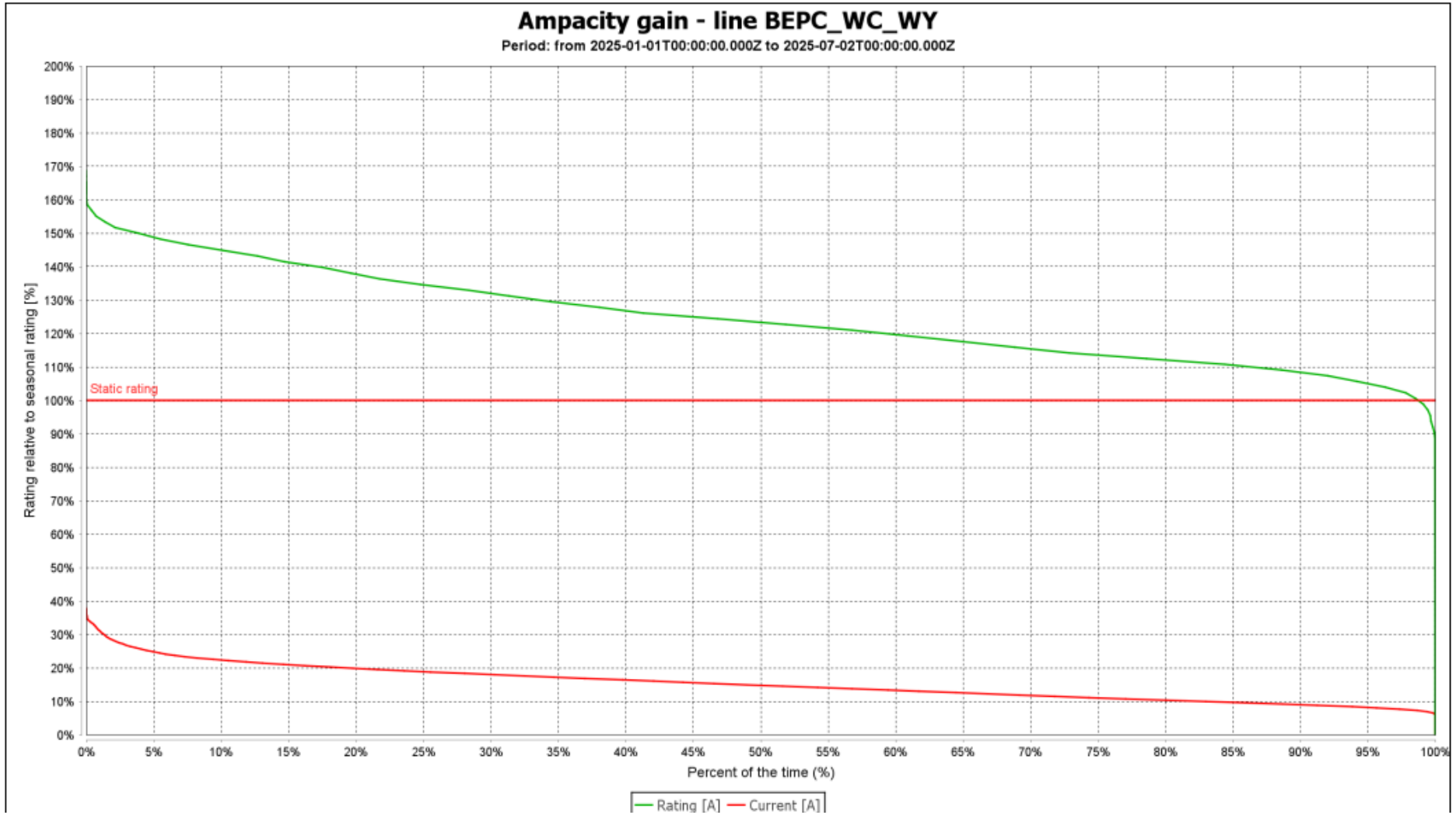
DLR Implementation: Between July & December 2024

- Yielded capacity above static rating 90% of the time.
- Avoided 500 hours of congestion (12% of period).
- Paid back full capital cost after just 38 days of operation.
- Over $20M of realized savings from congestion in 2024 alone.

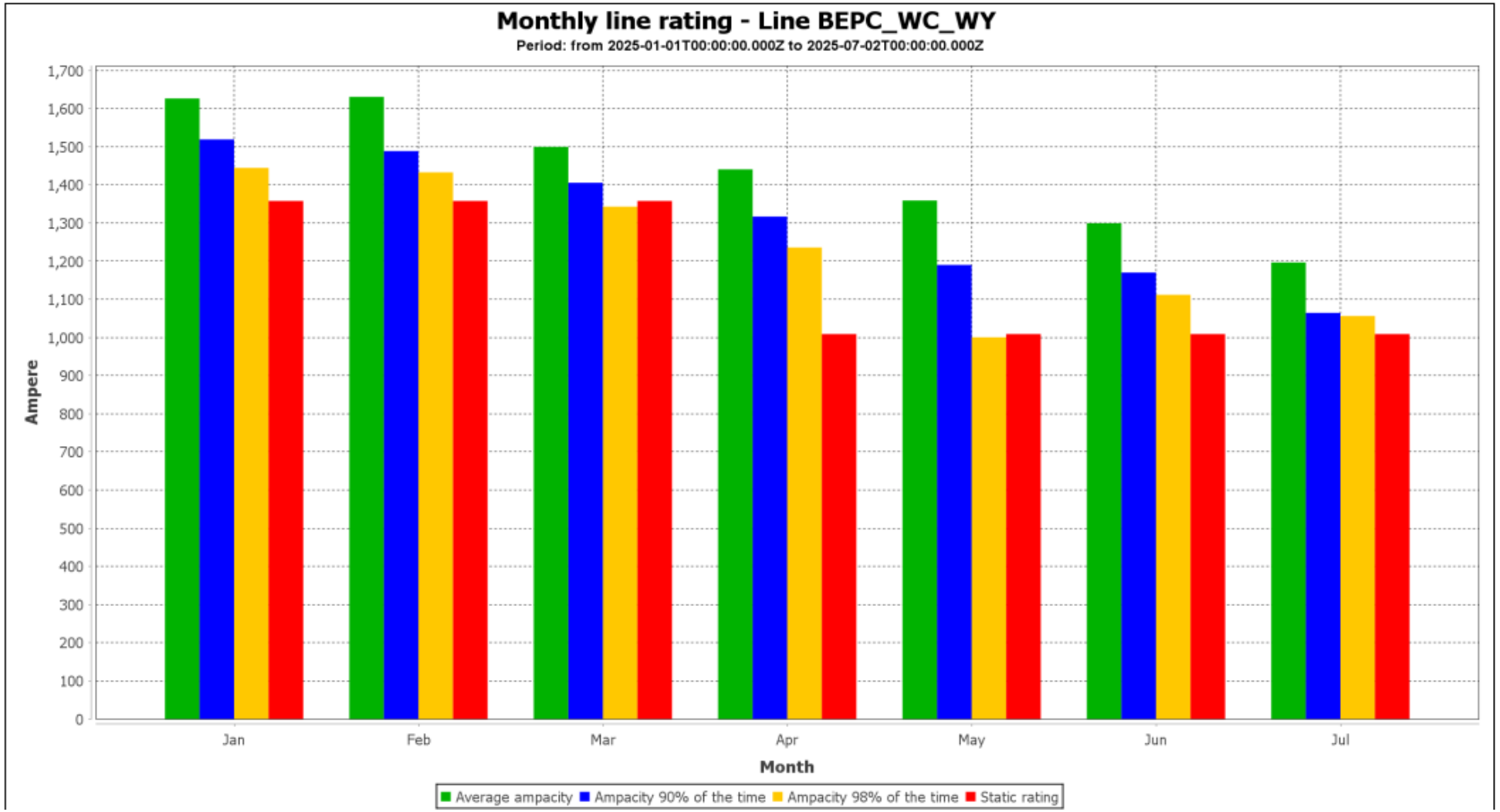
Other:

- Placed the Roundup to Kummer Ridge 345 kV line in service on December 17th 2024.

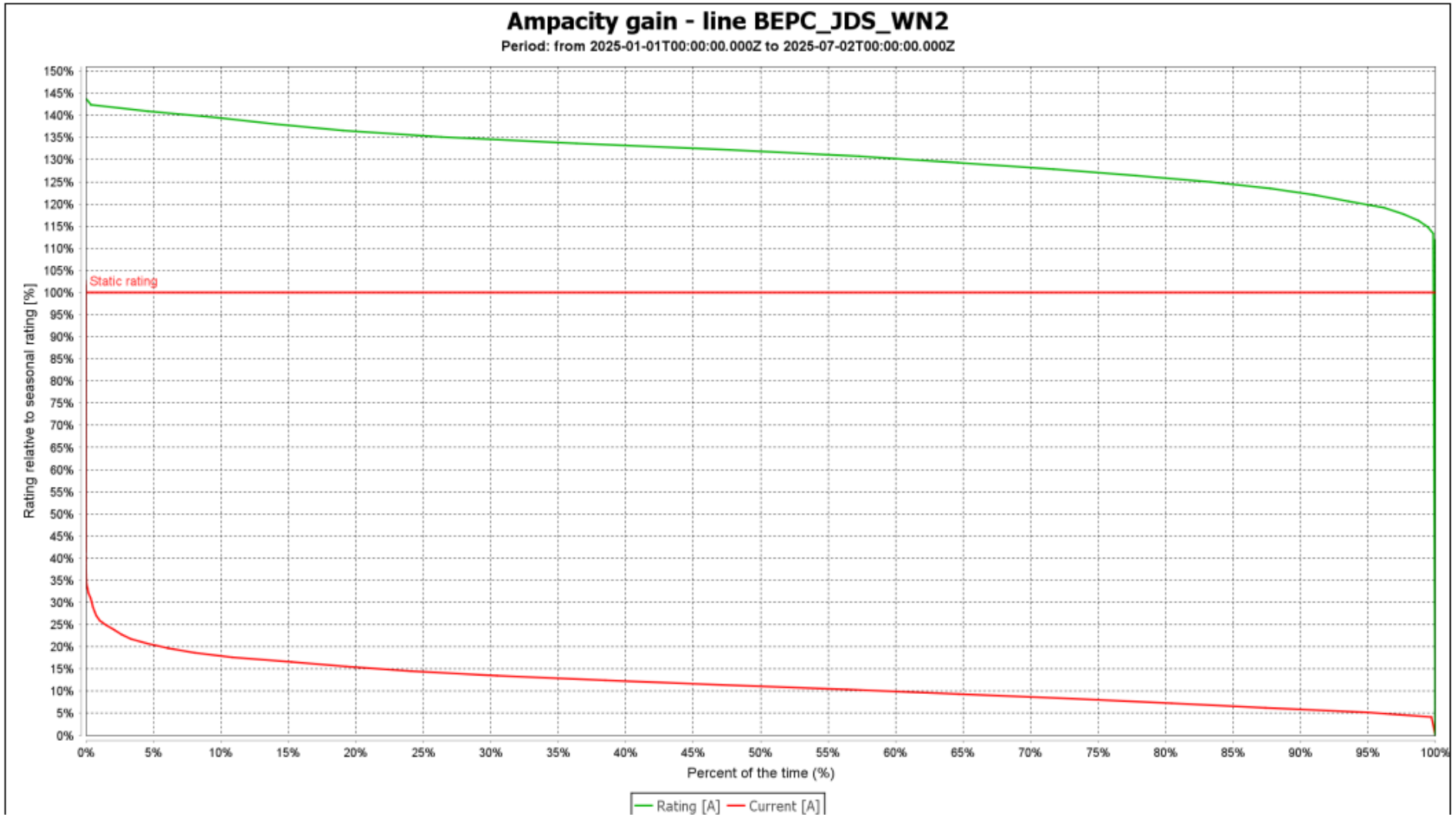
Line 1 Example – Ampacity Gained



Line 1 Example – Monthly Line Rating



Line 2 Example – Ampacity Gained



Questions

XI. Appendix B: Southwest Power Pool, Inc., Docket No. ER26- 2592-000 Submission of Tariff Revisions to Implement Economic Topology Optimization

May 21, 2026

The Honorable Debbie-Anne A. Reese
Secretary
Federal Energy Regulatory Commission
888 First Street, N.E.
Washington, D.C. 20426

Re: *Southwest Power Pool, Inc.*, Docket No. ER26-_____-000
Submission of Tariff Revisions to Implement Economic Topology Optimization

Dear Secretary Reese:

Pursuant to Section 205 of the Federal Power Act[1] ("FPA") and Part 35 of the Regulations of the Federal Energy Regulatory Commission ("Commission" or "FERC"),[2] Southwest Power Pool, Inc. ("SPP"), as authorized by its independent Board of Directors, submits revisions to Attachment AE of SPP's Open Access Transmission Tariff ("Tariff")[3] to allow for economic topology reconfigurations.

SPP is requesting that the Commission issue an order in this docket as soon as practicable, but not later than August 19, 2026, which is approximately ninety (90) days after submission of this filing. SPP requests an effective date of October 1, 2026, for the Tariff Records submitted in this filing in order to allow SPP staff to develop and test the proposed revisions.

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1 16 U.S.C. § 824d.

2 18 C.F.R. § 35.13.

3 Southwest Power Pool, Inc., Open Access Transmission Tariff, Sixth Revised Volume No. 1. References in this filing to "Tariff" refer to the version of SPP's Tariff currently in effect. "Proposed Tariff" refers to a version reflecting the revisions proposed in this filing. All capitalized terms not otherwise defined in this filing shall have the definitions assigned by the Tariff.

I. STAKEHOLDER APPROVAL

The proposed revisions were reviewed through the SPP stakeholder process, including meetings of the: Market Working Group ("MWG") on February 24, 2026[4] and March 24, 2026;[5] Regional Tariff Working Group ("RTWG") on February 26, 2026;[6] Operating Reliability Working Group on March 4, 2026,[7] and March 18, 2026;[8] and Markets and Operations Policy Committee ("MOPC") on April 14, 2026.[9] The SPP

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4 *See* MWG Meeting Minutes, February 24-25, 2026, at Agenda Item 8 posted at: https://www.spp.org/documents/76127/mwg%20meeting%20minutes%2020260224-25.pdf. The MWG develops and oversees policies and procedures related to the Integrated Marketplace protocols that define SPP's wholesale markets, including energy and operating reserve, congestion management, congestion hedging, demand response and market power mitigation. The MWG proposes changes to the Tariff and other governing documents to implement suggested market changes.

5 *See* MWG Meeting Minutes, March 24-25, 2026, at Agenda Item 9 posted at: https://www.spp.org/documents/76462/mwg%20meeting%20minutes%2020260325-26.pdf.

6 *See* RTWG Minutes, dated February 26, 2026, at Agenda Item 8c posted at: https://www.spp.org/documents/76143/rtwg%20minutes%2020260226.pdf. The RTWG develops, implements and oversees SPP's Tariff. The RTWG provides input on regulatory or implementation issues not specifically covered by the tariff and issues where there may be conflicting or differing interpretations of the tariff.

7 *See* ORWG Minutes, dated March 4-5, 2026, at Agenda Item 5 posted at: https://www.spp.org/spp-documents-filings/?id=19845 (see ORWG Meeting Materials 20260409 at file named 3a_ORWG_Minutes_20260304.docx). The ORWG develops and oversees policies and procedures related to the reliable and secure operation of the Bulk Electric System within SPP's Reliability Coordinator, Balancing Authority and Reserve Sharing Group footprints. The ORWG ensures these operating policies are consistent with North American Electric Reliability Corporation and regional reliability standards.

8 *See* ORWG Minutes, dated March 18, 2026, at Agenda Item 3 posted at: https://www.spp.org/spp-documents-filings/?id=19845 (see ORWG Meeting Materials 20260409 at file named 3a_ORWG_Minutes_20260318.docx).

9 *See* MOPC Minutes, dated April 14, 2026, at Agenda Item 3 posted at: https://www.spp.org/documents/76537/mopc%20minutes%20and%20ballots_

Board of Directors approved the proposed revisions at its meeting on May 5, 2026.[10] While SPP recognizes that stakeholder approval does not by itself cause a filing to be just and reasonable, SPP requests that the Commission extend appropriate deference to the wishes of SPP's stakeholders, consistent with Commission precedent.[11]

II. DISCUSSION OF PROPOSED REVISIONS

Traditionally, congestion management treats the transmission grid as "fixed," meaning that the preferred approach to managing congestion is to "redispatch," *i.e.*, to adjust resources' output to avoid or resolve congestion. Redispatch can result in increased production and congestion costs. However, in the absence of capable technology, the fixed topology assumption has been a necessary one because, historically, "finding

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combined%20file_v2.pdf. The MOPC consists of a representative officer or employee from each SPP Member and reports to the SPP Board of Directors. Its responsibilities include recommending modifications to the Tariff. *See* Southwest Power Pool, Inc., Bylaws, First Revised Volume No. 4 ("Bylaws") at Section 6.1.

[10] Board of Directors and Members Committee Meeting Minutes, dated May 5, 2026, at Agenda Item 3.8.1 posted at: https://www.spp.org/documents/76709/2026-05-05%20board%20of%20directors%20members%20committee%20meetings%20minutes.pdf.

[11] The Commission has previously recognized that provisions approved through RTO stakeholder processes are due deference. *See Sw. Power Pool, Inc.*, 127 FERC ¶ 61,283, at P 33 (2009) (noting that the Commission "accord[s] an appropriate degree of deference to RTO stakeholder processes"); *New Eng. Power Pool*, 105 FERC ¶ 61,300, at P 34 (2003) (Commission approval of transmission cost allocation proposal based upon an extensive and thorough stakeholder process); *Policy Statement Regarding Regional Transmission Groups*, 1991-1996 FERC Stats. & Regs., Regs. Preambles ¶ 30,976, at 30,872 (1993) (the Commission will afford the appropriate degree of deference to the stakeholder approval process). The Commission's deference to RTO stakeholder processes has been upheld by the courts. *See* Pub. Serv. Comm'n of Wis. v. FERC, 545 F.3d 1058, 1062-63 (D.C. Cir. 2008) (noting that the Commission often gives weight to RTO proposals that reflect the position of the majority of the RTO's stakeholders) (quoting *Am. Elec. Power Serv. Corp. v. Midwest Indep. Transmission Sys. Operator, Inc.*, 122 FERC ¶ 61,083, at P 172 (2008)).

optimization solutions in the necessarily real-time timelines [can be] extremely difficult…."[12]

That said, recently, the industry has found itself in a new age where technological advances, particularly grid-enhancing technologies ("GETs"), allow grid operators and experts to explore and embrace the "computational complexity"[13] of topology optimization. By "topology optimization," SPP refers to alteration of a transmission network's configuration in a way that economically and reliably manages congestion. Such reconfiguration has the potential to reduce production and congestions costs. By way of example, from 2020 to 2021, SPP congestion costs increased from $450 million to $1.2 billion.[14] GETs which allow for, say, topology reconfiguration have the potential to decrease costs such as these.

The revisions proposed herein effectuate a supplemental approach to congestion management. These revisions, which, again, SPP terms "topology optimization," aim to improve the efficiency and reliability of the grid by economically and reliably altering the network configuration. Market Participants may submit proposals for reconfigurations due to outages or constraints, or SPP may itself proffer a reconfiguration. In either case, SPP then analyzes the potential reconfiguration to determine whether it satisfies the requirements set forth in the Tariff, including, for example, whether the market as a whole will benefit from the reconfiguration. As SPP does with other processes today, SPP works with the affected local transmission operators ("TOP") to mitigate any reliability risk posed by the reconfiguration. If none are identified, and as real-time conditions allow, then SPP will implement the solution.

This proposal is not novel. For example, in a February 2022 report, the Department of Energy noted,

> Transmission topology optimization is a software technology that identifies reconfigurations in the grid to route power flow around congested or overloaded

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12 *Building for the Future Through Electric Regional Transmission Planning and Cost Allocation*, Order No. 1920, 187 FERC ¶ 61,068, P 1246, *order on reh'g and clarification*, Order No. 1920-A, 189 FERC ¶ 61,126 (2024), *order on reh'g and clarification*, Order No. 1920-B, 191 FERC ¶ 61,026 (2025).

13 *Id*.

14 Southwest Power Pool Market Monitoring Unit, State of the Market 2021 (May 10, 2022), at 3, available at: https://www.spp.org/documents/67104/2021%20annual%20state%20of%20the%20market%20report.pdf.

> transmission elements, taking advantage of the meshed nature of the bulk-power grid. The reconfigurations are implemented by switching high voltage circuit breakers. By more evenly distributing flow over the network, topology optimization increases the transfer capacity of the grid.[15]

More recently, in June 2025, Potomac Economics published a report, writing,

> *Transmission Reconfiguration*. It can often be highly economic to alter the configuration of the network (e.g., opening a breaker) to reduce flows on a severely constrained transmission facility. This is done currently to mitigate reliability concerns under procedures established with the transmission owners impacted by the reconfiguration. Such procedures should be expanded to economically manage congestion as they can mitigate or eliminate severe congestion on some constraints.
>
> MISO has established a process to evaluate potential reconfiguration options proposed by participants. We recommend MISO develop a tool to identify such options itself.[16]

Indeed, in June 2023, the Midcontinent Independent System Operator, Inc. ("MISO") established a formal reconfiguration process with stakeholders, becoming the first Regional Transmission Organization ("RTO")/Independent System Operator ("ISO") to allow market participants to request grid reconfigurations to reduce regional

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15 U.S. Department of Energy, Grid-Enhancing Technologies: A Case Study on Ratepayer Impact (Feb. 2022), at 8, available at: https://www.energy.gov/sites/default/files/2022-04/Grid%20Enhancing%20Technologies%20-%20A%20Case%20Study%20on%20Ratepayer%20Impact%20-%20February%202022%20CLEAN%20as%20of%20032322.pdf.

16 Potomac Economics, 2024 State of the Market Report for the MISO Electricity Markets (June 2025), at Executive Summary, available at: https://www.potomaceconomics.com/wp-content/uploads/2025/06/2024-MISO-SOM_Report_Body_Final.pdf.

congestion.[17] Thirty-two (32) reconfigurations were submitted in 2024 and 2025, nine (9) of which were implemented.[18] Likewise, the Electric Reliability Council of Texas, Inc. ("ERCOT"), which currently utilizes topology optimization software to support its Constraint Management Plan, has set implementation of its own topology optimization mechanism for the first quarter of 2027.[19]

Recently, a 2024 analysis assessed the potential for GETs to facilitate transmission tools (including topology optimization) in the PJM Interconnection, L.L.C. ("PJM") footprint.[20] The study found that topology optimization could reduce the cost to alleviate 72 overloads by $273 million by using software to determine alternate grid configurations that reroute power around constraints, such as opening or closing breakers automatically. The report also stated GETs can significantly reduce the amount of time to make the necessary grid adjustments to bring new generation online, addressing concerns PJM has raised about the balance of deactivations and new resource entry, as well as reducing energy costs by speeding development of low-cost renewables. The report estimated ratepayers could save $1.1 billion in annual production costs by 2033 against a $0.1 billion installation cost for GETs. Although these findings applied to PJM specifically, they offer a compelling case for the widespread deployment of GETs, including topology optimization. Indeed, PJM already utilizes "topology control" within its footprint, meaning it uses a topology control software to identify the benefits of transmission switching, and to make recommendations accordingly to alleviate constraints.[21]

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17 MISO RSC Meeting, MISO Economic Reconfiguration Process Value and Suggestions for Further Improvements (Oct. 9, 2025), available at: https://cdn.misoenergy.org/20251009%20RSC%20Item%2011%20Reconfiguration%20for%20Cost%20Congestion%20Process%20Enhancements721553.pdf.

18 *Id*. at Slide 4.

19 ERCOT, Grid Transformation – Technology Initiatives - Roadmaps, available at: https://www.ercot.com/files/docs/2025/08/29/GRIT-roadmap.pdf (indicating ERCOT plans to implement topology optimization in the first quarter of 2027).

20 RMI, GETting Interconneciton in PJM, Grid-Enhancing Technologies Can Increase the Speed and Scale of New Entry from PJM's Queue (February 2024), available at: https://rmi.org/wp-content/uploads/dlm_uploads/2024/02/GETs_insight_brief_v3.pdf.

21 PJM, Innovation & Grid Enhancing Technologies at PJM (Oct. 21, 2024), available at: https://opsi.us/wp-content/uploads/2024/10/3.-20241021-Bryson-OPSI-Innovation-Slides-Session-3.pdf.

And, in January 2025, the Electric Power Research Institute ("EPRI") published a whitepaper entitled, "Transmission Topology Optimization, a State-of-the-Art White Paper," writing,

> This fixed-topology assumption is not due to a perspective that transmission switching actions are not useful in maintaining reliability and managing congestion. Indeed, some control centers have for decades used such actions where they can be identified easily and pose no additional risk, e.g., opening an overloaded low-capacity circuit to eliminate the overload. Rather, the fixed-topology assumption is motivated by the fact that switching actions beneficial to grid reliability and congestion management that pose no additional risk have in the past been difficult to identify; this is because the number of options is very large, requiring deployment of computationally intensive methods to identify them.
>
> More recently, optimization algorithms have improved, and additional techniques have been demonstrated. At the same time, the prevalence of nodal electricity markets and their identification of congestion costs has provided a convenient means of quantifying potential economic and reliability benefits obtained from switching actions at scale (which we will also refer to as network reconfiguration). As a result, there has been a maturation of the software technology known today as transmission topology optimization (TTO). TTO provides the ability to efficiently identify beneficial and safe switching actions in grid operation for managing network security and congestion.
>
> …
>
> Perhaps the best indication of industry's high interest in TTO lies in the number of studies RTOs have conducted recently to assess TTO's viability and benefits. This

> includes pilot projects or case studies at PJM[22], SPP,[23] and ERCOT.[24] Results from the studies conducted by NewGrid and SPP show that topology control can achieve $18–44 million in market cost savings annually.[25]

Indeed, as EPRI describes, case studies regarding the efficacy of topology optimization in SPP's own market have demonstrated the benefits.

In 2018, SPP conducted an extended pilot with a topology optimization software tool, with a reliability focus on finding "preferred" solutions consisting of a single switching action at or below 230 kV with at least 5% N-1 loading reduction, less than 30 MW of newly radialized load, and no resultant constraint loading over 95% post-contingent. SPP engineers evaluated 100 flowgates with congestion during real-time operations and found 'preferred' solutions to 55 flowgates. Some of these reconfigurations were then implemented in real-time operations.[26]

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22 P. A. Ruiz, M. Caramanis, E. Goldis, X. Li, K. Patel, R. Philbrick, A. Rudkevich, R. Tabors, and B. Tsuchida, *Transmission Topology Optimization* (June 28, 2016), available at: https://cms.ferc.gov/sites/default/files/2020-05/20160629114654-2%2520-%2520PRuiz%2520FERCTechConf%252028Jun2016_FINAL_2.pdf.

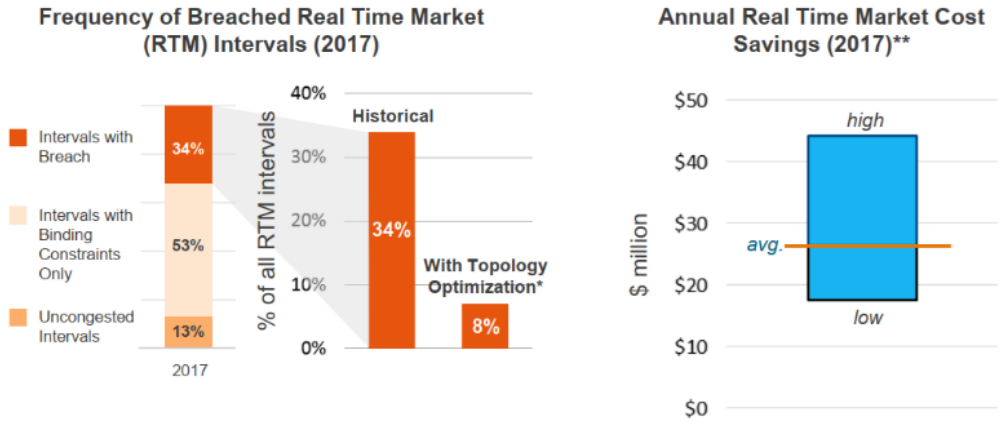
23 P. A. Ruiz and J. Caspary, *SPP Transmission Topology Optimization Pilot* (March 20 2019), available at: https://watt-transmission.org/wp-content/uploads/2019/03/spp-transmission-topology-optimization-pilot-efficient-congestion-management-and-overload-mitigation-through-system-reconfigurations-.pdf.

24 P. A. Ruiz, J. Caspary, and L. Butler, *Transmission Topology Optimization Case Studies in SPP and ERCOT*, FERC Tech Conf on Increasing Day-Ahead and Real-Time Market Efficiency and Enhancing Resilience through Improved Software, Docket No. AD10-12-011 (June 24, 2020), available at: https://www.ferc.gov/sites/default/files/2020-06/W3-1_Ruiz_et_al.pdf.

25 EPRI, *Transmission Topology Optimization, State-of-the-Art White Paper: A GET SET White Paper* (January 16, 2025), at 3, 9 available at: https://www.epri.com/research/products/000000003002031445.

26 Pablo Ruiz, Jay Caspary and Luke Butler, *Transmission Topology Optimization Case Studies in SPP and ERCOT*, FERC Tech Conf on Increasing Day-Ahead and Real-Time Market Efficiency and Enhancing Resilience through Improved Software, Docket No. AD10-12-011 (June 24, 2020), at Slides 12-20, available at: https://www.ferc.gov/sites/default/files/2020-06/W3-1_Ruiz_et_al.pdf.

Continuing this analysis, NewGrid, SPP, and the Brattle Group evaluated the utilization of topology optimization to identify reconfigurations of major constraints binding or overloading. The evaluation found that historical system operating limit violations could be eliminated for 75% of the constraints analyzed. The resulting congestion cost savings in the real-time markets were estimated to be in the range of $18-44 million annually.[27]



NewGrid, SPP, and the Brattle Group also evaluated the potential to use topology optimization to develop Corrective Action Plans ("CAPs") that relied on reconfigurations and avoided load shedding during certain North American Electric Reliability Corporation (NERC)-defined events. SPP's planning department selected three events for evaluation, one of each NERC type listed, that had CAPs consisting of load shedding (up to 243 MW of load shed). The reconfiguration solutions identified

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[27] Pablo A. Ruiz, Doug Bowman, Kathryn Dial, Xiao Li, Ryan Schoppe, Zachary Sharp, Jason Terhune, Bruce Tsuchida, *Transmission Topology Optimization Applications to Enhance System Reliability and Resilience and to Increase Market and Planning Efficiency in Southwest Power Pool*, FERC Tech Conf on Increasing Day-Ahead and Real-Time Market Efficiency and Enhancing Resilience through Improved Software, Docket No. AD10-12-009) (June 26, 2018), at Slides 13-21, available at: https://www.ferc.gov/sites/default/files/2020-08/T4-2_Ruiz.pdf; Pablo A. Ruiz, Doug Bowman, Kathryn Dial, Xiao Li, Ryan Schoppe, Zachary Sharp, Bruce Tsuchida, *Transmission Topology Optimization Pilot Study to Support Congestion Management and Ice Buildup Mitigation* (November 15, 2018), Presentation #6, Slides 16-20, available at: https://www.spp.org/Documents/59018/2018 spp tech expo materials.zip.

and validated by SPP provided full relief and eliminated the need to shed load in those CAPs.[28]

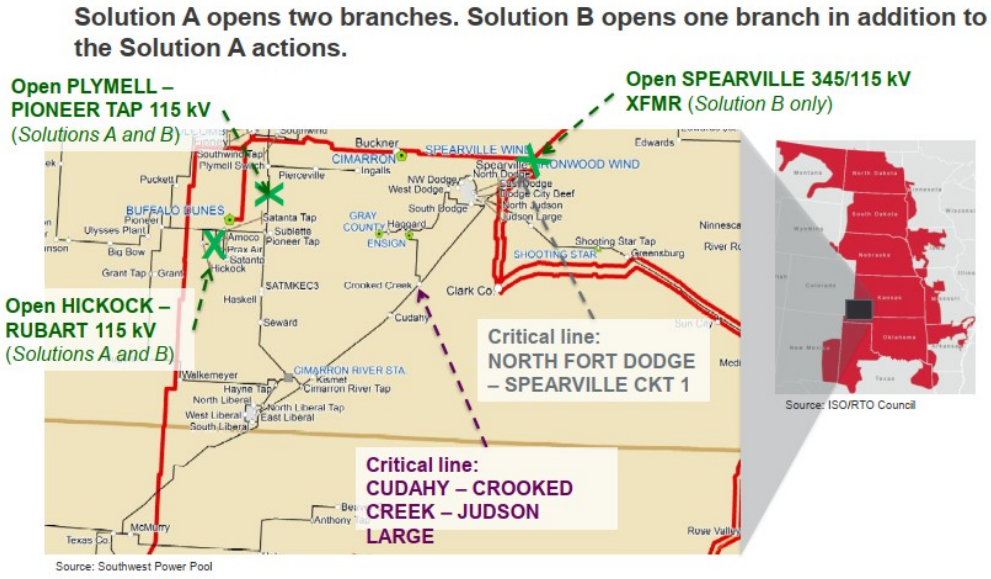
Case Study Type	Flow on Violated Branch Initial [% of Rating]	Flow on Violated Branch With Solution [% of Rating]	Avoided Load Loss [MW]	No. of Actions	No. of New Constraints >95% flow	No. of New Constraints >100% flow	No. of New Constraints <0.9 pu volt	Radialized Load [MW]
P6 Event	129%	86%	243	2	1	0	0	65
P7 Event	107%	94%	55	2	0	0	0	0
Extreme Event	113%	97%	151	1	0	0	0	0

NewGrid, SPP and The Brattle Group also evaluated the potential to use topology optimization during icing conditions to mitigate mechanical line failure risk. The analysis focused on reconfigurations to redirect flow to critical lines at risk of failure due to ice accumulation to prevent or limit ice formation. The study was performed for the January 2017 Winter Storm Jupiter conditions, which caused multiple transmission outages due to ice accumulation, and focused on the Sunflower Electric Power Corporation footprint. The study identified two reconfiguration solutions that could have significantly relieved the ice buildup on selected critical lines, while meeting reliability criteria. The estimated savings of avoiding the hypothetical outages of these critical lines were $10 to $17 million, in addition to avoided system restoration costs.[29]

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[28] Pablo A. Ruiz, Doug Bowman, Kathryn Dial, Xiao Li, Ryan Schoppe, Zachary Sharp, Jason Terhune, Bruce Tsuchida, *Transmission Topology Optimization Applications to Enhance System Reliability and Resilience and to Increase Market and Planning Efficiency in Southwest Power Pool*, FERC Tech Conf on Increasing Day-Ahead and Real-Time Market Efficiency and Enhancing Resilience through Improved Software, Docket No. AD10-12-009 (June 26, 2018), at Slides 23-24, available at: https://www.ferc.gov/sites/default/files/2020-08/T4-2_Ruiz.pdf.

[29] Pablo A. Ruiz, Doug Bowman, Kathryn Dial, Xiao Li, Ryan Schoppe, Zachary Sharp, Bruce Tsuchida, Transmission Topology Optimization Pilot Study to Support Congestion Management and Ice Buildup Mitigation (November 15, 2018), Presentation #6, Slides 22-32, available at: https://www.spp.org/Documents/59018/2018 spp tech expo materials.zip.



And, in 2022, Evergy, an SPP Market Participant, analyzed ten significant constraints in the Evergy footprint that are difficult to control and are thus prone to reliability challenges. Evergy identified reconfigurations that, had they been implemented, could have eliminated 98% of overloads and reduce congestion costs by 85% for congestion patterns associated with those constraints, including the impacts of additional constraints resulting from the change in flow patterns. This *ex-post* analysis was conducted using snapshots posted by SPP over a 4-month period in the spring and summer of 2022.[30]

During Winter Storm Elliott (December 23-24, 2022), heavy congestion limited supply to a major metropolitan area in the SPP footprint. SPP implemented two transmission reconfigurations that released up to 845 MW from otherwise stranded generation under Energy Emergency Alert conditions, which supplied 14 GWh of additional energy during the event. Two complementary reconfigurations identified but not used during the storm would have released up to another 600 MW of constrained-off generation to supply an additional 9 GWh during the event. Together, these reconfigurations could have avoided 36 hours of transmission constraints.[31]

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30 Pablo Ruiz and Derek Brown, *Reliable and Efficient Congestion Mitigation using Transmission Reconfigurations*, SPP Seams Advisory Group October 7, 2022 Meeting Materials at Agenda item 5, at Slides 19-21, available at: https://www.spp.org/spp-documents-filings/?id=20425.

31 Pablo Ruiz, *Topology Optimization for Least-Cost and Reliable Congestion Mitigation*, FERC Tech Conf on Increasing Market and Planning Efficiency through Improved Software, Docket No. AD10-12-014 (June 29, 2023), at Slide

Importantly, reconfigurations can be effective mitigation solutions until a transmission expansion project can be completed. NewGrid identified reconfigurations to mitigate two such major constraints in Oklahoma identified by SPP in its Integrated Transmission Planning ("ITP") process as transmission needs. The reconfigurations would have enabled 10-23% increases in flow throughput in the area.[32]

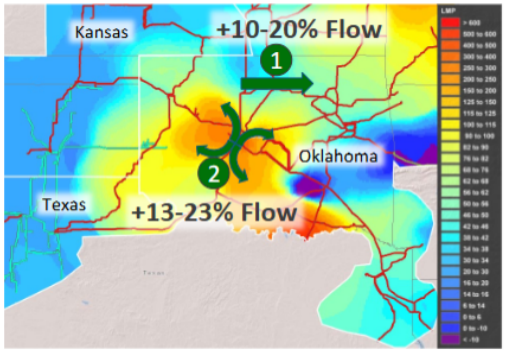
1 Osage – Webb Tap 138 kV flo Sooner – Cleveland 345 kV

- Constraint is very heavily loaded
  - Breaching or binding over 28% of all market intervals in April 2022
  - Active (high flows but not binding) over 60% of all market intervals in April 2022
- SPP identified the constraint as "overlapping Reliability and Economic need" in the 2020 ITP
- Reconfiguration enables 10-20% of increased flow

2 Cimarron XF 345/138 kV constraint

- Breaching or binding over 5% of all market intervals in April 2022, active (high flows, not binding) over 92% of such intervals
- Increases load costs in Oklahoma City (see contour map, the constraint was breaching at the time)
- SPP identified this constraint as the top "Operational need" in the 2020 ITP Assessment Report
- Reconfiguration reliably enables 13-23% increased constraint throughput under congested conditions

SPP Real Time Price Contour Map, June 21, 2022 18:50, https://pricecontourmap.spp.org/pricecontourmap/



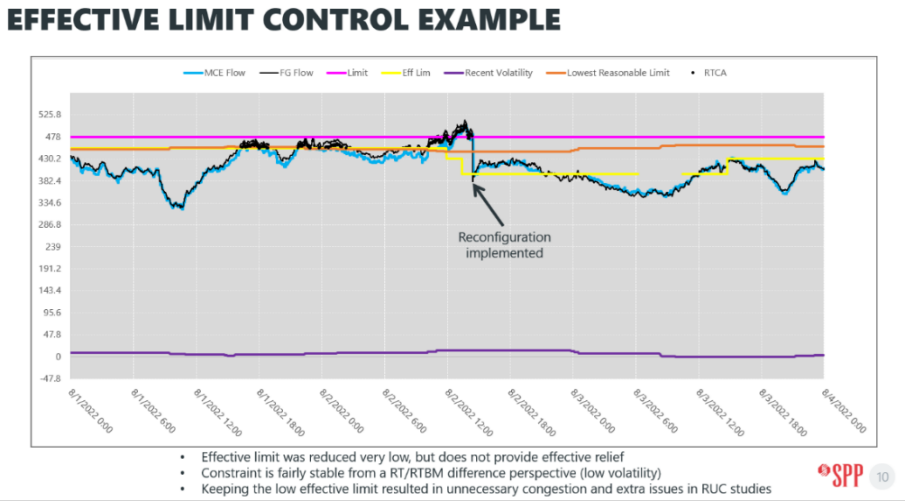
In its role as a reliability coordinator, SPP has experience implementing a select number of reconfigurations. For example, the top- and fourth-most expensive constraints in SPP in 2019 arose from contingency limitations on a 69 kV line in Oklahoma. These constraints were permanently resolved with a transmission reconfiguration. Prior to their resolution in 2020, these constraints were binding 16% of all Real-Time Balancing Market intervals in 2019, costing the market about $25 million that year. In addition to being costly and frequently binding, these constraints had resulted in almost $30 million of Transmission Congestion Rights ("TCR") Market underfunding.[33]

25, available at: https://www.ferc.gov/media/pablo-ruiz-newgrid-somerville-ma.

[32] FERC, *Congestion and Overload Mitigation Using Optimal Transmission Reconfiguration*, at Slide 15, available at: https://www.ferc.gov/media/congestion-and-overload-mitigation-using-optimal-transmission-reconfigurations-experience.

[33] Southwest Power Pool Market Monitoring Unit, State of the Market 2019 (May 11, 2020), at 194-195, 199 and 214, available at:

And, in 2022, one of the most significant constraints in SPP was a transformer providing a transmission connection for a major metropolitan area. Under high load conditions, especially with medium to high wind, this constraint could overload due to limited generation options to manage it. During the summer of 2022, SPP implemented a reconfiguration to fully mitigate this congestion when the constraint was overloading and uncontrollable with redispatch. The reconfiguration provided approximately 20% reduction in flow for this constraint.[34]



With the above-described policies, discussions, and analyses in mind, SPP proposes to add new Section 4.6 of Attachment AE to the Tariff:

> **4.6 Economic Topology Optimization**
>
> Economic topology optimization analyses assess the impacts of reconfiguring the system topology to relieve congestion such that market production cost is reduced while maintaining reliability.

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https://www.spp.org/documents/62150/2019%20annual%20state%20of%20the%20market%20report.pdf.

34 Kathryn Dial, *Flowgate Metrics & Congestion Management Discussion,* SPP Operating Reliability Working Group September 28-29, 2022, Meeting Materials, at Agenda Item 9, Slides 9-10, available at: https://www.spp.org/spp-documents-filings/?id=19845.

> A Market Participant may submit an economic topology optimization request to the Transmission Provider, or the Transmission Provider may itself develop economic topology optimization scenarios for analysis. The following information must be included in any submission, which is further described in the Market Protocols:
>
> 1. Target permanent or temporary flowgate;
> 2. Operational action specifications;
> 3. Expected duration of operational action specifications; and
> 4. Expected conditions of operational action specifications.
>
> The Transmission Provider may set a limit to the number of economic topology optimization requests which may be submitted by Market Participants and to the number of economic topology optimization requests that the Transmission Provider will study during any calendar month.[35]

As set forth above, either a Market Participant may request a topology reconfiguration, or SPP itself may develop such a scenario. Market Participants' option to submit requests incentivizes Market Participants to suggest configurations which SPP staff, whose time is not fully dedicated to scouring the system for economic reconfigurations, may not otherwise have the opportunity to identify.

In the event that a Market Participant requests an optimization, the Market Participant must identify in its request the flowgates targeted with the request, as well as the operational action specifications (including the expected duration and the expected conditions). Without this information, SPP would be unable to analyze the request.

Finally, within Section 4.6, SPP proposes that a limit may be set to the number of requests submitted or studied for the purpose of ensuring that SPP staff, at least initially, is not overwhelmed by requests or studies to the point where studying requests becomes infeasible. Given that this process is new to SPP, there exists some level of uncertainty as to the appetite Market Participants may have for making requests, as well as to the time it will take SPP's staff to conduct its studies. The Market Protocols provide additional details as to the limits described in the Tariff.

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[35] Proposed Tariff, Attachment AE, Section 4.6.

In addition to Section 4.6 ("Economic Topology Optimization"), SPP proposes to add a new Section 4.6.1, entitled, "Economic Topology Optimization Analysis," as follows:

> **4.6.1 Economic Topology Optimization Analysis**
>
> For valid requests, the Transmission Provider will analyze economic topology optimization requests under multiple scenarios, based on the expected state of the system at the time the reconfiguration is expected to be in place. In order for an economic topology optimization request to pass the Transmission Provider's analysis, the request must meet the Transmission Provider's requirements for each of the following:
> 1. System reliability
> 2. Regional production cost; and
> 3. Transmission Congestion Rights Market funding
>
> The Transmission Provider will reject the request if any one of the considerations do not meet the Transmission Provider's requirements.[36]

First, Section 4.6.1 states that SPP, as Transmission Provider, will analyze only "valid" requests. By "valid," the language refers to (1) those requests which satisfy the requirements set forth in Section 4.6, meaning those requests which submit the necessary information; and (2) requests which are not in exceedance of any limit described in the Market Protocols.

Second, Section 4.6.1 describes the analysis undertaken by SPP. Specifically, SPP's analysis will assess the proposed reconfiguration's effect on (1) system reliability; (2) regional production cost; and (3) TCR Market funding. If the request does not satisfy any one of these enumerated points, then SPP will deny the request.

First, with respect to system reliability, if SPP's analysis demonstrates that the reconfiguration would detrimentally affect the system's reliability, then the request will be denied. This is consistent with how SPP takes market actions today—no market action is undertaken if the market action will interfere with system reliability.

Second, with respect to "regional production cost," any reconfiguration must result in production cost savings, meaning that lower costs are realized as a result of the reconfiguration by, for example, reduced congestion costs or the displacement of

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[36] Proposed Tariff, Attachment AE, Section 4.6.1.

higher-cost generation.[37] This criterium is significant because it means that no single Market Participant's request can be granted unless the market as a whole sees benefit in the request reconfiguration.

Third, with respect to TCR Market funding, SPP will ensure that, to the extent the reconfiguration may affect TCR Market funding, any such effects are outweighed by production cost benefit.[38] Otherwise, any effects on the congestion hedging market could defeat the purpose of reconfiguration.

Finally, SPP proposes to add a new Section 4.6.2 ("Economic Topology Optimization Results"):

> **4.6.2 Economic Topology Optimization Results**
>
> If a request passes the Transmission Provider's analysis, the Transmission Provider will coordinate with the reconfigured facilities' local transmission operator(s), which will perform a local reliability assessment and inform the Transmission Provider if the request is approved or denied. The Transmission Provider may re-evaluate the need for any approved economic topology optimization requests as system conditions change and, as appropriate, may (a) revert the system topology to pre-request state; or (b) adjust the system topology to accommodate the system conditions change.[39]

A TOP, as defined by NERC, is the "entity responsible for the reliability of its 'local' transmission system, and that operates or directs the operations of the transmission Facilities." NERC Reliability Standards require the TOP to maintain the reliability of its transmission and facilities.[40] For example, NERC Reliability Standard TOP-001-6,

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37 The Market Protocols describe this metric in greater detail. *See* Revision Request 747 Recommendation Report ("RR 747"), Market Protocols, Section 4.2.8.3.1(2) ("Economic Analysis"), available at: https://www.spp.org/Documents/75922/RR747.zip.

38 The Market Protocols describe this metric in greater detail. *See* RR 747, Market Protocols, Section 4.2.8.3.1(3) ("Economic Analysis").

39 Proposed Tariff, Attachment AE, Section 4.6.2.

40 The Federal Energy Regulatory Commission reviews, approves, and enforces mandatory reliability standards developed by NERC. Reliability Explainer, https://www.ferc.gov/reliability-explainer.

Requirement 1 states, "Each Transmission Operator shall act to maintain the reliability of its Transmission Operator Area via its own actions or by issuing Operating Instructions."

Accordingly, the TOP has long been responsible for maintaining the reliability of its local system, which is within SPP's footprint. With the revisions proposed in Section 4.6.2, SPP simply maintains its existing relationship with the TOP. Specifically, SPP will request that the TOP evaluate a proposed reconfiguration consistent with that TOP's existing responsibility under NERC Reliability Standards. As a matter of course, and in the same way that SPP as a reliability coordinator *must* study the reliability impacts of a proposed reconfiguration, the TOP *must* study the reliability impacts of the same with respect to its system.[41]

Additionally, within Section 4.6.2, SPP adds language which contemplates re-evaluation of any approved optimizations. This may become necessary because as the time of the reconfiguration timeline approaches, the request could have become moot or even harmful if, say, a planned outage assumed in the original analysis had already occurred or got cancelled. Therefore, the language gives SPP the ability to revert the system or to adjust the optimization.

III. EFFECTIVE DATE AND REQUEST FOR WAIVER

SPP requests an effective date of October 1, 2026, for the proposed revisions, which is more than 120 days after the submission of this filing. Therefore, SPP requests waiver of the Commission's notice requirements.[42] Good cause exists for the proposed revisions to be effective on October 1, 2026, in accordance with the Commission's waiver of notice requirement codified in Section 35.11 of the Commission's regulations[43] because SPP will need a number of months to develop, test, and implement the software system changes for the Tariff revisions proposed herein. SPP requests the Commission issue an order by August 19, 2026, to allow SPP sufficient time to make the necessary preparations prior to implementation.

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41 Any such study would be well-documented by the TOP. *See*, *e.g.*, TOP-001-6, Measure 1 ("Each Transmission Operator shall have and provide evidence which may include but is not limited to dated operator logs, dated records, dated and time-stamped voice recordings or dated transcripts of voice recordings, electronic communications, or equivalent documentation, that will be used to determine that it acted to maintain the reliability of its Transmission Operator Area via its own actions or by issuing Operating Instructions.").

42 18 C.F.R. § 35.3(a)(1).

43 18 C.F.R. § 35.11.

IV. ADDITIONAL INFORMATION

A. Information Provided Per Commission Regulations[44]

1. Documents submitted with this filing:
In addition to this Transmittal Letter, Clean and Redlined Tariff revisions under the Sixth Revised Volume No. 1.

2. Service:
SPP has electronically served a copy of this filing on all its Members, Transmission Customers, and Market Participants. A complete copy of this filing will be posted on the SPP web site, www.spp.org, and is also being served on all affected state commissions.

3. Requisite agreements:
Not Applicable.

B. Communications

Correspondence and communications regarding this filing should be sent to, and SPP requests that the Secretary include on the official service list, the following:

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[44] Because the revisions to the Tariff submitted herein do not involve any changes in rates, the use of the abbreviated filing procedures set forth in 18 C.F.R. § 35.13(a)(2)(iii) is appropriate.

V. CONCLUSION

For all of the foregoing reasons, SPP respectfully requests that the Commission issue an order accepting the Tariff revisions proposed herein as just and reasonable, to be effective on October 1, 2026, and issue an order not later than August 19, 2026.

Respectfully submitted,

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